

Bellman-sufficient Information Complexity

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Abstract

We develop Bellman-sufficient information complexity, a formal representation-level framework for sequential decision making. The primitive benchmark is a fixed-truth environment space Ω with unrestricted nonanticipating algorithms. The intrinsic object is a Bellman-sufficient state representation, serving as an interactive notion of sufficient statistics, together with an information index $Y = \chi(\Omega)$, often the optimal decision or value object rather than the full environment. On the upper-bound side, learning is organized as a dynamic program on the sufficient state, equipped with a logarithmic information potential for the index. On the lower-bound side, a Bellman–Fano method uses the same state representation and information index, but propagates separate Bellman recursions for information gain and ghost mass. The central matching statement is therefore a conditional Bellman information-risk sandwich: when the log-penalized Bellman upper value and the ghost-quantile lower value close at the same radius, they certify the same complexity scale. Popular algorithms then appear as tractable relaxations of this common log-potential Bellman program, rather than as separate notions of information complexity.

Contents

1	Introduction	2
1.1	Main contributions.	4
1.2	Organization.	5
2	Sequential decision making with Bellman-sufficient representations	6
2.1	Sequential decision making and Bellman state compression	6
2.2	Information index, conditional loss, and entropy accounting	9
2.3	Bellman-sufficient representation	10
2.4	Basic examples of Bellman-sufficient representations	12
2.5	Full environment posterior is fallback sufficient state	13
2.6	Why sufficient states and compression matter	14
3	Information complexity sandwich	16
3.1	Code length, hard priors, and ghost entropy	16
3.2	From entropy comparison to regret comparison	18
4	Indexed information and exact AIR/MAIR identities	21
4.1	Posterior-reference histories and index information	21
4.2	Fixed-truth indexed AIR bracket	22
4.3	Exact AIR identity and model-index specialization	23

5	Upper bounds: one identity and four algorithm families	27
5.1	Information-potential Bellman programming	30
5.2	UCB families: calibration plus optimism	32
5.3	E2D: robust one-step offset optimization	33
5.4	AMS/EBO: robust convex belief optimization	33
6	Lower bounds: reference histories and quantile indices	35
6.1	Ghost probability and true good probability	35
6.2	Reference-history quantile theorem	36
6.3	Bellman–Fano lower-bound method	37
6.4	DEC as one-step complexity	39
7	Applications: information-complexity sandwich examples	40
7.1	Finite-action kernel bandits: algorithms and upper bounds	40
7.2	Gaussian multi-armed bandits: finite-index sandwich matching	46
7.3	Hypercube linear bandits: high-dimensional indexed sandwich	49
8	Conclusion	54

1 Introduction

Information-theoretic minimax theory asks how much risk remains when an experiment can reveal only limited information. In classical noninteractive estimation the experiment is fixed before the data are observed, and sharp rates are obtained by matching an upper information-risk construction with a lower entropy calculation: KL information measures what the experiment can distinguish, while local prior mass or packing entropy measures how many alternatives remain statistically indistinguishable. This is the perspective behind the information-risk upper and lower bounds of [Zhang \(2006\)](#), the entropy characterization of minimax rates in [Yang and Barron \(1999\)](#), and the classical Fano–Assouad method ([Yu, 1997](#)). In interactive decision making and dynamic control, the learner’s actions change the experiment itself. The learner chooses actions by a nonanticipating policy, so the experiment itself is adapted to the past. The main challenge is therefore that decisions, observations, and information accumulation are not separate objects; they are produced jointly by a single controlled stochastic process.

Algorithmic Fano’s method is one way to respect this adaptivity. It compares the real history with an algorithm-dependent reference, or “ghost,” history rather than with a fixed nonadaptive sampling scheme. This viewpoint appears in recent interactive Fano frameworks for interactive decision making ([Chen et al., 2024](#)) and in algorithmic upper and lower bounds for representation learning ([Li and Xu, 2026](#); [Xu, 2026](#)). It is also closely related to the decision-estimation coefficient (DEC) approach of [Foster et al. \(2021\)](#) and its constrained or localized refinements ([Foster et al., 2023](#)), as well as to the earlier information-ratio approach ([Russo and Van Roy, 2014](#); [Lattimore and György, 2021](#)). One central difficulty is that a sharp lower bound may not collapse a T -round adaptive trajectory into a one-round comparison too early. Without proper localization, such a collapse can erase the geometry of the evolving posterior, confidence set, or reference state, precisely the geometry on which a matching upper bound may rely.

A separate motivation, and a broader challenge, is that existing information-theoretic minimax theory is not yet well suited to dynamic control and planning across episodes ([Silver and Sutton, 2025](#); [Liu et al., 2025](#); [Maiti et al., 2026](#)). The present paper suggests that closing gaps in episodic

interactive decision making and developing theory for across-episode dynamic control share a common missing diagnostic: Bellman recursion and dynamic programming should appear explicitly on both the upper- and lower-bound sides. We frame this perspective within a unified sequential decision-making framework.

The thesis of this paper is that the intrinsic object for sequential decision making is an indexed Bellman-sufficient representation. The frequentist environment space Ω , together with the unrestricted nonanticipating benchmark, identifies the primitive sequential decision-making objective. A Bellman-sufficient state representation turns the problem into a dynamic program over a sufficient state: a sufficient statistic, model posterior, frequentist estimator, or other representation that summarizes the environmental and historical information needed to close the Bellman recursion. The index $Y = \chi(\Omega)$ specifies the information target whose acquisition is charged. It may be the optimal action, an optimal policy, a value object, an active finite marginal, or the full environment.

The same formalism also explains why information potentials are not limited to statistical bandits. In planning, search, or reasoning, the latent instance may be a world model, problem instance, proof environment, or task specification, while the index may be a plan, answer, policy, value function, or proof object. Whenever the process admits a Bellman-sufficient state and a maintained marginal on that index, the logarithmic mass potential is again the generic information potential. This paper keeps most of the technical development in statistical bandits and reinforcement learning, where the upper and lower bounds can be stated cleanly.

When a sufficient state representation closes the Bellman recursion with a smaller information index, the learner need not pay for estimating irrelevant features of the model. This is the sense in which learning is a special dynamic program: unlike ordinary computational dynamic programming, where the value function is problem-specific, the statistical potential is generically logarithmic in the index belief. The exact log-penalized Bellman program is the upper object that can match the Bellman–Fano ghost lower certificate; popular algorithms including UCB, E2D, and AMS/EBO are tractable relaxations or certificates of this same dynamic information principle.

The basic identity on the upper-bound side is formulated at a fixed frequentist truth and expressed through Bellman recursion and the chosen information index. Fix a truth ω^* , let $y^* = \chi(\omega^*)$, and let q_t be the retained reference marginal on the index. The specialized logarithmic potential

$$\phi_t(s; y^*) = \gamma[-\log q_t(y^*)]$$

turns the one-step fixed-truth log gain into a Bellman information potential. The corresponding indexed Algorithmic Information Ratio (AIR) bracket is

$$\ell_{\omega^*}(S_t, p_t) + \mathbb{E}_{\omega^*}[\phi_{t+1}(S_{t+1}; y^*) \mid S_t, p_t] - \phi_t(S_t; y^*),$$

and its sum telescopes to a regret bound of order $\gamma \log(1/q_1(y^*))$ plus the accumulated bracket errors (Xu and Zeevi, 2025). Bayesian posterior averaging of the same coordinate identity gives the chain rule

$$I_\mu(Y; H_T) = \mathbb{E}_{H'_T} \sum_{t=1}^T \mathcal{I}_\chi(S'_t, p'_t),$$

where S'_t is the reference posterior state and p'_t is the algorithm’s action distribution at that reference history. The key algorithmic principle is that the logarithmic mass penalty should be propagated through a controlled Bellman recursion.

The lower-bound side is organized by the Bellman–Fano method. The index specifies which uncertainty is charged, and the reference history specifies which low-regret ghost trajectories are counted. The algorithm-specific quantile theorem, Theorem 6.1, is an application of the interactive

Fano method (Chen et al., 2024) to Bellman-sufficient representations; Theorem 6.3 is its algorithm-uniform Bellman-recursion certificate. In its simplest algorithm-specific form one obtains

$$\text{if } T\bar{C}_\chi^{\text{Alg}}(\mu) \leq \text{kl}\left(\frac{1}{2}, p_r^{\text{Alg}}(\mu, \chi)\right), \quad \text{then } \mathbb{E}_{\Omega \sim \mu} \mathbb{E}_{\mathbb{P}_\Omega^{\text{Alg}}} L_\Omega(H_T) \geq \frac{Tr}{2}.$$

Here r is an average-regret threshold, p_r^{Alg} is the posterior-reference ghost probability of average regret at most r , and $\bar{C}_\chi^{\text{Alg}} = T^{-1} \mathbb{E} \sum_t \mathcal{I}_\chi(S'_t, p'_t)$ is average indexed information. The critical balance is

$$T\bar{C}_\chi \asymp \log \frac{1}{p_r}.$$

When the ghost-good mass is roughly $\exp(-d_{\text{eff}})$, the admissible total information is d_{eff} -level. This is the Fano/local-entropy radius, not the constant two-point radius. Theorem 6.3 converts the algorithmic lower-bound approach into a frequentist minimax statement. The Bellman–Fano method may use the exact Bellman-recursion values for information capacity and ghost mass, or valid supersolutions that upper-bound those exact values.

Theorem 3.8 states the resulting minimax information-risk sandwich. After a state, index, reference update, and calibration mechanism have been fixed, the upper side pays a log-penalized Bellman value with coordinate cost $\log(1/q_1(\chi(\omega)))$; the lower side pays a Bellman–Fano ghost entropy $\log(1/p_r)$. The central comparison is therefore the objective ratio between these two logarithmic quantities. When the ghost entropy has the same order as the upper code length, and the Bellman upper value has regular growth in its information budget, the upper and lower regret bounds match at the same radius. This is the interactive analogue of noninteractive information-risk matching: fixed-design local prior mass is replaced by ghost-good mass along an adaptive reference history, and fixed-sample KL is replaced by a controlled Bellman information telescope. We use “Bellman-sufficient information complexity” for this compatible state/index accounting. It becomes a matching complexity statement only after the upper and lower certificates are separately verified and shown to agree up to the stated constants or logarithmic factors.

The word “Bellman” in the title is deliberate. The exact value of an interactive problem is a dynamic program, following Bellman’s principle of optimality (Bellman, 1957). An upper bound is a Bellman supersolution, or admissible relaxation in the sense of Rakhlin et al. (2012): it lies above the dynamic value and gives an algorithm. A lower bound is an exact Bellman–Fano value comparison, or a computable Bellman–Fano certificate obtained by upper-bounding the exact information-capacity and ghost-good-mass values. The canonical information-theoretic upper algorithm is therefore the AIR/MAIR information-potential Bellman program itself: it keeps the continuation value on the indexed state and optimizes the Bellman bracket before any one-step relaxation is imposed. UCB (Auer et al., 2002; Abbasi-Yadkori et al., 2011; Srinivas et al., 2010), E2D (Foster et al., 2021, 2023), and AMS/EBO (Lattimore and György, 2021; Foster et al., 2022; Xu and Zeevi, 2025; Liu et al., 2025, 2026) are useful because they certify, relax, or robustify this Bellman bracket; they should not be confused with the exact Bellman program itself.

1.1 Main contributions.

1. We formulate *sequential decision making with Bellman-sufficient representations*, a formal representation-level framework for reinforcement learning, optimal control, and interactive decision making. In this framework, one studies the unrestricted nonanticipating benchmark through Bellman recursion, while a chosen index $Y = \chi(\Omega)$ specifies the decision-relevant information whose acquisition is charged. The state must close feasible actions, fixed-truth prediction, loss evaluation, and updating; when posterior-reference information is used, it must

also close posterior predictive and conditional-index laws. Several bandit and reinforcement learning examples illustrate the framework with states given by sufficient statistics, model posteriors, frequentist estimators, or other Bellman-sufficient representations.

2. We prove the indexed posterior-reference chain rule and the generic indexed AIR regret identity. AIR is recovered by taking the decision index $Y = A^*(\Omega)$, while MAIR is recovered by taking the model or environment index $Y = \Omega$. We then record simplified statements and alternative proofs, show how the gradient bracket unifies different update rules, and explain the role of Danskin’s theorem in the original AIR proof of [Xu and Zeevi \(2025\)](#).
3. We isolate the main information-complexity sandwich. The upper value pays an initial coordinate code length, while the Bellman–Fano lower value is governed by ghost entropy. We prove entropy-gap and regret-gap comparison statements under regularity conditions.
4. We give a unified “one identity, four algorithm families” account centered on a log-penalized Bellman upper theorem. The first family is the exact logarithmic information-potential Bellman program; this is the tightest upper object that can match the Bellman–Fano lower value. UCB methods certify the fixed-truth log-potential bracket through self-normalized calibration and optimism. E2D methods optimize a one-step offset obtained by replacing the continuation value by a local statistical separation penalty. AMS/EBO methods optimize KL-regularized robust belief relaxations, equivalently dual log-partition relaxations, of the same indexed AIR bracket.
5. We formulate the algorithmic quantile lower theorem and the Bellman–Fano method as algorithm-uniform minimax lower bounds. The lower-bound argument uses the same stepwise indexed information, state representation, and Bellman recursion to control both information gain and ghost mass. Bandit examples show that this approach recovers matching lower bounds.
6. We organize the examples as information-complexity sandwich tests. Kernel bandits illustrate the upper bounds and the need for localization. Gaussian multi-armed bandits give a complete finite-index matching example. Hypercube linear bandits give a high-dimensional indexed lower bound with standard upper bounds matching up to logarithmic factors.

1.2 Organization.

Section 2 defines sequential decision making, information indices, and Bellman-sufficient representations, and records the basic sufficient-statistic and full-posterior examples. Section 3 states the information-complexity sandwich and the entropy/regret comparison principles. Section 4 gives the indexed information telescope and the exact AIR/MAIR regret identities. Section 5 gives the log-penalized Bellman upper theorem and upper algorithmic relaxations including UCB families, E2D, and AMS/EBO. Section 6 gives the quantile lower theorem and the Bellman–Fano method. Section 7 is organized as a sequence of sandwich examples: kernel-bandit upper bounds, Gaussian multi-armed-bandit matching, and hypercube linear-bandit matching.

2 Sequential decision making with Bellman-sufficient representations

2.1 Sequential decision making and Bellman state compression

The primitive object is sequential decision making on histories. The framework encompasses both bandits and reinforcement learning in rich environments (Lattimore and Szepesvári, 2020; Sutton and Barto, 2018), as well as planning, search, reasoning, and optimal control (Silver and Sutton, 2025; Cormen et al., 2001; Bertsekas, 2017). The central point is simple: from basic online-learning models to dynamic planning and reasoning models, future losses, interactions, and observations generally depend on the history (Cesa-Bianchi and Lugosi, 2006; Recht, 2019; Chen et al., 2021; Hazan et al., 2025). The sequential decision-making model considered here is designed to reflect this dependence.

A formal model of the sequential decision-making problem. There is an environment class Ω , and performance is required for every fixed truth $\omega^* \in \Omega$. A prior μ is introduced only when we form a Bayes value, a Yao lower bound, a posterior-reference trajectory, or an algorithmic belief. The primitive history used in the paper is the full pre-action history, denoted H_{t-1} . To avoid separate notation for contexts, physical states, public randomization, and protocol variables, we regard observations as packets. There may be an initial packet O_0 before the first decision, and recursively

$$H_0 = O_0, \quad H_t = (H_{t-1}, A_t, O_t).$$

The packet O_t contains whatever is publicly revealed after the round- t action and before the next decision: feedback, the next context, the next physical state, the next stage marker, public randomization, or other protocol information. Thus the object called the “current context” in a contextual bandit is part of O_{t-1} , hence part of H_{t-1} , before A_t is chosen. The learner’s decision rule is not part of the history; it is the policy being optimized. Private randomization may be represented by the decision kernel, or conditioned on when analyzing a fixed randomized implementation.

At time t , after $h \in \mathcal{H}_{t-1}$, the learner may choose an action in a measurable set $\mathcal{A}_t(h)$. An unrestricted nonanticipating algorithm is a sequence of kernels

$$\pi_t^{\text{Alg}}(\cdot | h) \in \Delta(\mathcal{A}_t(h)), \quad h \in \mathcal{H}_{t-1},$$

with deterministic algorithms included as degenerate kernels. Let $\mathfrak{A}_T^{\text{na}}$ denote the class of all such algorithms, and write

$$p_t = \pi_t^{\text{Alg}}(\cdot | H_{t-1}), \quad A_t \sim p_t.$$

The most general one-step primitives allow the environment or an adaptive adversary to depend on the announced mixed action as a current control:

$$O_t \sim P_{\omega^*, t}(\cdot | H_{t-1}, p_t, A_t), \quad \ell_t(\omega^*, H_{t-1}, p_t, A_t). \quad (1)$$

When the law and loss depend only on the realized action, as in ordinary stochastic bandits and Markov decision processes, we suppress the argument p_t and write $P_{\omega, t}(\cdot | H_{t-1}, A_t)$ and $\ell_t(\omega, H_{t-1}, A_t)$. The increment may be signed, provided the cumulative target used in a lower-bound theorem is integrable and has the stated lower bound, usually $L_\omega(H_T) \geq 0$. The Bellman clock t is abstract. In episodic reinforcement learning it may represent a macro-episode, or the micro-time pair consisting of an outer episode and an inner horizon stage.

The exact frequentist objective in this paper is

$$\mathfrak{R}_T^*(\Omega) := \inf_{\text{Alg} \in \mathfrak{A}_T^{\text{na}}} \sup_{\omega \in \Omega} \mathbb{E}_{\omega}^{\text{Alg}} \sum_{t=1}^T \ell_t(\omega, H_{t-1}, p_t, A_t).$$

This is the reinforcement-learning and interactive-decision-making problem in its general sequential risk-minimization form: choose a nonanticipating algorithm whose Bellman risk is small uniformly over environments.

Bellman state and Bellman sufficiency. Our theory centers on the specification of an analytical Bellman state for sequential decision making. This object should not be confused with the physical state often used in Markov decision process models. A compressed Bellman state is a dynamic sufficient statistic for decision making,

$$S_t = \phi_t(H_{t-1}).$$

Because the observation packets already contain contexts, physical states, public randomization, and other pre-action variables, no separate raw-history notation is needed. If an implementation retains an additional memory object, that object is either a measurable function of the history, included in the observation packets, or conditioned on in the analysis. Such a compression may be used in Bellman recursions only if the available actions, observation law, loss, and update are measurable with respect to the retained state and the current control. Thus there must exist an action set $\mathcal{A}_t(s)$, a kernel $P_{\omega,t,s,p,a}$, and a loss $\ell_{\omega,t}(s,p,a)$ such that, whenever $S_t(h) = s$,

$$\begin{aligned} \mathcal{A}_t(h) &= \mathcal{A}_t(s), \\ P_{\omega,t}(\cdot \mid h, p, a) &= P_{\omega,t,s,p,a}, \\ \ell_t(\omega, h, p, a) &= \ell_{\omega,t}(s, p, a). \end{aligned}$$

When the environment does not observe the announced mixed action, we suppress p and write $P_{\omega,t,s,a}$ and $\ell_{\omega,t}(s,a)$. There must also be an update map or controlled transition, written in the general notation as

$$S_{t+1} = \tau_t(S_t, p_t, A_t, O_t),$$

and shortened to $\tau_t(S_t, A_t, O_t)$ when p_t is irrelevant. If any of these conditions fails, the retained state is incomplete and must be augmented; see Definition 2.3 for the formal statement. The full history $S_t = H_{t-1}$ always satisfies the fixed-truth Bellman closure, because all current contexts, physical states, public randomization, and other pre-action variables have been included in the observation-packet history. As illustrated in Section 2.5, a full environment posterior together with the current time/context/physical state also provides a fallback sufficient state for broad classes of Bayesian reference experiments.

The notation $P_{\omega,t,s,a}$ and $\ell_{\omega,t}(s,a)$ means only that all observed variables needed to determine the next observation law and the loss have been included in the Bellman state s . In applications this state may include an estimator or posterior object, the relevant physical state, an adversary state, or a relaxation variable. It does not mean that a bandit arm or an MDP transition kernel literally depends on an estimator alone. Once a Bellman-sufficient state is fixed, all main Bellman operators optimize over $p \in \Delta(\mathcal{A}_t(s))$. In stationary examples, or when the stage t is included in s , we sometimes write $\mathcal{A}(s)$ for $\mathcal{A}_t(s)$.

Reference dynamic program and minimax lower bound. On a sufficient Bellman state, with update $S_{t+1} = \tau_t(S_t, p_t, A_t, O_t)$, or with p_t suppressed when irrelevant, the Bayes/reference dynamic program induced by a posterior b_s is

$$V_{T+1}(s) = 0, \quad V_t(s) = \inf_{p \in \Delta(\mathcal{A}_t(s))} \left\{ \ell(s, p) + \mathbb{E}_{a \sim p, o \sim P_{s,p,a}} V_{t+1}(\tau_t(s, p, a, o)) \right\}, \quad (2)$$

where

$$\ell(s, p) = \mathbb{E}_{a \sim p} \int \ell_\omega(s, p, a) b_s(d\omega), \quad P_{s,p,a} = \int P_{\omega,s,p,a} b_s(d\omega).$$

Equation (2) is not the definition of the frequentist problem; it is the Bellman recursion induced by a chosen reference representation. Minimax lower bounds are connected to such reference recursions through Yao’s principle,

$$\mathfrak{R}_T^*(\Omega) \geq \sup_{\mu \in \Delta(\Omega)} \inf_{\text{Alg} \in \mathfrak{A}_T^{\text{na}}} \mathbb{E}_{\Omega \sim \mu} \mathbb{E}_{\mathbb{P}_\Omega^{\text{Alg}}} \sum_{t=1}^T \ell_t(\Omega, H_{t-1}, p_t, A_t).$$

This offers a lower-bound perspective on information complexity at the level of broad environment classes, rather than only for particular classes.

Remark 2.1 (Adversarial and estimated losses). *The sequential decision-making framework naturally accommodates general adversarial losses beyond the stochastic regret setting (Cesa-Bianchi and Lugosi, 2006; Abernethy et al., 2011). A nonstochastic or adaptive adversary may be treated as part of the fixed environment ω , provided its rule is nonanticipating. Classical online learning often permits the adversary to observe the history as well as the learner’s announced mixed action. Then the mixed action is the current control, not an additional component of the history being optimized over, and the general primitives in (1) are*

$$\ell_t(\omega, H_{t-1}, p, a), \quad P_{\omega,t}(\cdot \mid H_{t-1}, p, a).$$

Equivalently, after p is chosen one may view (H_{t-1}, p) as the post-announcement pre-sampling information; the Bellman operator must still optimize over p . With the full history as state and p as a control argument, this representation is automatic for any fixed nonanticipating adversary rule. Estimated-loss upper bounds can be handled by replacing the true loss in the Bellman bracket with a state-measurable surrogate or domination certificate, while carrying the resulting approximation error. Lower bounds require the target cumulative indexed loss used in the ghost event to be lower bounded, usually nonnegative after a harmless shift.

Remark 2.2 (Sublinear cumulative loss against the optimal dynamic policy). *Related finite-time minimax and information-theoretic phenomena appear in sequential decision-making domains where the cumulative loss, measured against an optimal dynamic policy, can be sublinear without an explicit episodic restarting structure. Examples include linear quadratic control (Lai, 1986; Mania et al., 2019; Simchowitz and Foster, 2020; Jedra and Proutiere, 2022), blind network revenue management (Besbes and Zeevi, 2012), bandits with knapsacks (Badanidiyuru et al., 2018), and, more recently, queueing theory and stochastic networks (Liu et al., 2025; Liang et al., 2026). These settings are important in their own right and are different in flavor from standard finite-horizon or discounted reinforcement-learning formulations, where the planning horizon or discount factor often appears explicitly in regret bounds (Azar et al., 2017; Shah et al., 2022). Existing results provide valuable model-specific upper and lower bounds, but they are not yet part of a general Bellman-recursive information-complexity theory. A central direction is to understand whether the representation-level machinery developed here can recover and extend such bounds for broader dynamic programming, planning, search, and reasoning problems.*

2.2 Information index, conditional loss, and entropy accounting

The latent environment space is Ω , but the information charged by the upper and lower bounds may be a coarser coordinate. An information index is a measurable map

$$Y = \chi(\Omega) \in \mathcal{Y}.$$

The index is chosen by the analyst and should reflect the decision-relevant object: an optimal arm, an optimal policy, a value function, a finite active marginal, or, when no compression is justified, the full environment. The fixed-truth Bellman recursion still uses ω^* . The index enters through the logarithmic potential, the posterior/reference information accounting, and the lower ghost event.

The loss need not be a function of Y alone. For Bayes or Fano calculations we therefore use the conditional index loss, assuming the relevant regular conditional laws exist. In the general convention with a current mixed-action control,

$$\ell_t^\chi(y, h, p, a) := \mathbb{E}_\mu[\ell_t(\Omega, h, p, a) \mid \chi(\Omega) = y, H_{t-1} = h]. \quad (3)$$

When the one-step primitives do not depend on p , the argument is suppressed. When the retained state is an exact posterior-indexed lift in Definition 2.3, this conditional loss and the corresponding conditional predictive law are state-measurable and may be written as $\ell_{t,y}^\chi(s, p, a)$ and $P_{t,s,p,a}^y$. These conditional-index objects are not required for ordinary fixed-truth Bellman control; they are required for indexed posterior averaging and for the Bellman–Fano lower-bound method.

We use the cumulative and average losses, with the same suppression convention,

$$L_\omega(H_T) = \sum_{t=1}^T \ell_t(\omega, H_{t-1}, p_t, A_t), \quad \bar{L}_\omega(H_T) = T^{-1} L_\omega(H_T),$$

and

$$L_\chi(y, H_T) = \sum_{t=1}^T \ell_t^\chi(y, H_{t-1}, p_t, A_t), \quad \bar{L}_\chi(y, H_T) = T^{-1} L_\chi(y, H_T).$$

Under the Bayesian mixture generated by $\Omega \sim \mu$,

$$\mathbb{E}_{\Omega \sim \mu, H_T \sim \mathbb{P}_\Omega^{\text{Alg}}} L_\Omega(H_T) = \mathbb{E}_{Y, H_T} L_\chi(Y, H_T). \quad (4)$$

For the upper identities, the increments may be any integrable costs for which the displayed expectations exist. For the quantile lower bounds, the object that must be nonnegative is the cumulative indexed loss, not necessarily each one-step increment. Throughout the clean lower-bound statements we assume $L_\chi(Y, H_T) \geq 0$ almost surely under the true mixture. If a problem has a known lower bound $L_\chi \geq -B$, the same statements apply to the shifted loss $L_\chi + B$, with the corresponding shift subtracted at the end.

The two logarithmic quantities compared later are the upper coordinate code length and the lower ghost entropy. If q_1 is the initial reference marginal on $\mathcal{Y}$, the fixed-truth upper telescope pays

$$L_0(\Omega_0; q_1, \chi) := \sup_{\omega \in \Omega_0} \log \frac{1}{q_1(\chi(\omega))}.$$

If q_1 is uniform on a finite index set of size M , then $L_0 = \log M$. At a regret radius r , the Bellman–Fano lower theorem uses the ghost-good probability

$$p_r = \mathbb{P}_{Y \sim q_1, H'_T} \{\bar{L}_\chi(Y, H'_T) \leq r\}, \quad E_r := \log \frac{1}{p_r}.$$

Thus the primitive entropy comparison is L_0/E_r , not the size of the full model class. The detailed comparison and its conversion into a regret gap are stated in Section 3.

2.3 Bellman-sufficient representation

We now give the formal representation axioms used throughout the rest of the paper. The frequentist truth ω^* remains fixed. Priors, posteriors, confidence objects, exponential-weights distributions, and algorithmic beliefs are reference objects: they may define algorithms or certificates, but they are not substitutes for the fixed environment unless a theorem explicitly takes a Bayesian average.

Definition 2.3 (Indexed Bellman-sufficient representation). *Fix an index map $\chi : \Omega \rightarrow \mathcal{Y}$. A retained state process*

$$S_t = \phi_t(H_{t-1})$$

is a fixed-truth Bellman-sufficient representation if the following objects are determined by the current time-state pair (t, s) .

- (i) *Admissible actions. There is a state-measurable action set $\mathcal{A}_t(s)$ such that $\mathcal{A}_t(h) = \mathcal{A}_t(s)$ whenever $S_t(h) = s$.*
- (ii) *Predictive sufficiency. For every environment ω , mixed action $p \in \Delta(\mathcal{A}_t(s))$, and realized action $a \in \mathcal{A}_t(s)$, there is a kernel $P_{\omega,t,s,p,a}$ such that, for every history h with $S_t(h) = s$,*

$$P_{\omega,t}(\cdot \mid h, p, a) = P_{\omega,t,s,p,a}.$$

When the environment does not react to the announced mixed action, the argument p is suppressed.

- (iii) *Loss sufficiency. There is an integrable state-measurable cost function $\ell_{\omega,t}(s, p, a) \in \mathbb{R}$ such that*

$$\ell_t(\omega, h, p, a) = \ell_{\omega,t}(s, p, a) \quad \text{whenever } S_t(h) = s.$$

The lower-bound quantile theorem does not require every increment to be nonnegative; it requires the cumulative indexed loss to be lower bounded, and in the displayed lower bounds we assume $L_\chi(Y, H_T) \geq 0$ almost surely.

- (iv) *Update sufficiency. There is an update map or controlled kernel τ_t such that the next retained state is generated from the retained state and the current interaction. In deterministic-update notation,*

$$S_{t+1} = \tau_t(S_t, p_t, A_t, O_t),$$

with p_t suppressed when irrelevant. Random next contexts, physical next states, stage markers, and public randomization are part of the observation packet O_t .

When these clauses hold, Bellman recursion is closed under each fixed truth. A compressed state is therefore a dynamic sufficient statistic for the Bellman primitives: feasible actions, prediction, loss evaluation, and continuation.

If, in addition, a prior or reference law μ is fixed and one wants exact state-based posterior-reference information accounting, the full-history posterior objects must descend to the state. On the full history define

$$b_{t,h} := \mathcal{L}_\mu(\Omega \mid H_{t-1} = h), \quad q_{t,h} := \chi_{\#} b_{t,h},$$

and, for $q_{t,h}$ -almost every y ,

$$b_{t,h}^y := \mathcal{L}_\mu(\Omega \mid H_{t-1} = h, \chi(\Omega) = y).$$

The full-history index-conditional predictive law and loss are

$$P_{t,h,p,a}^y := \int P_{\omega,t}(\cdot \mid h, p, a) b_{t,h}^y(d\omega),$$

$$\ell_{t,h,p,a}^x(y) := \int \ell_t(\omega, h, p, a) b_{t,h}^y(d\omega).$$

An exact posterior-indexed lift on the state is the additional fiber-invariance requirement that, whenever $S_t(h) = s$, there exist state-measurable objects satisfying

$$q_{t,h} = q_{t,s}, \tag{5}$$

and, for $q_{t,s}$ -almost every y ,

$$P_{t,h,p,a}^y = P_{t,s,p,a}^y, \quad \ell_{t,h,p,a}^x(y) = \ell_{t,y}^x(s, p, a), \tag{6}$$

together with a state-measurable reference update for $q_{t,s}$. These posterior-predictive and conditional-index clauses are not needed for the fixed-truth Bellman recursion itself. They are needed when the posterior-averaged chain rule, the Bayes information-potential program, or the Bellman–Fano lower certificate is run on the compressed state.

The notation will usually suppress the explicit time subscript when the stage is part of the state, and it will suppress the mixed-action argument when the environment does not react to the announced distribution. Thus $P_{\omega,s,a}$, $\ell_{\omega}(s, a)$, and $\mathcal{A}(s)$ mean the appropriate special cases of $P_{\omega,t,s,p,a}$, $\ell_{\omega,t}(s, p, a)$, and $\mathcal{A}_t(s)$ at the current Bellman time. This convention is harmless only after the state and current control include the variables that make actions, losses, observation laws, and updates state-measurable.

Bellman sufficiency versus classical sufficiency and Bellman rank. Classical sufficiency provides the static prototype for Bellman-state sufficiency (Fisher, 1922): a statistic $T(X)$ is sufficient for a family $\{P_\theta\}$ when the conditional distribution of the data given T is independent of θ . Bellman sufficiency is a closely related but conceptually distinct controlled analogue. The retained state must preserve everything needed for future actions, predictions, losses, information coordinates, and updates under admissible controls. In particular, one need not estimate all of ω^* if a smaller state closes these recursions.

As explained above, Bellman sufficiency is a closedness condition for the controlled recursion, not a requirement of classical exact sufficiency expressed through conditional independence. The exact state-based lift in Definition 2.3, namely the fifth axiom in equations (5)–(6), is therefore a state-level requirement and should be distinguished from full-history posterior sufficiency. Full-history posterior objects are tautological on H_{t-1} . A compressed state satisfies the exact indexed lift only when the quantities needed by the indexed Bellman recursion are constant on the fibers of S_t : the indexed marginal, the index-conditional predictive law, and the index-conditional loss. Thus full model-posterior sufficiency implies the lift, but the lift can be weaker. It should be read as an indexed conditional-moment closure condition: the state is sufficient for the conditional Bellman experiment given the index, not necessarily for the full environment; see Recht (2019, Section 3.3) for a possible connection between this viewpoint and the classical reinforcement-learning and control literature. This distinction is important for Section 2.6, which explains why sufficient states matter and why they can be tighter, or more intrinsic, than using the full history without compression.

The index $Y = \chi(\Omega)$ is the coordinate whose logarithmic mass is charged by the upper information potential and whose ghost-good mass appears in the lower Fano certificate. It need not

identify the whole environment. In bandits it may be the optimal arm; in reinforcement learning it may be an optimal policy or value object; in planning or reasoning it may be a plan, proof, or answer. The mathematics is the same whenever a Bellman-sufficient state and a reference marginal q_s for the index are available.

The formalism is aligned with the representation-level philosophy behind low Bellman rank (Jiang et al., 2017), but it is deliberately stricter. Bellman rank is a factorization of expected Bellman errors relative to a function class, roll-in distribution, and witness family. It is not, by itself, a state: it does not determine the realized next observation, the fixed-truth loss, the posterior or reference update, or the conditional laws for an index. It becomes an indexed Bellman-sufficient representation only after those missing objects are supplied and shown to close the recursions above.

The following basic examples belong in the framework section because they verify Definition 2.3 directly. They also fix the interpretation of the state used later: the fixed truth remains the environment, while sufficient statistics, posteriors, and finite marginals are retained Bellman states or reference objects.

2.4 Basic examples of Bellman-sufficient representations

Structured bandit problems offer particularly transparent examples of the Bellman-sufficiency framework for sequential decision making, which is guided by the classical intuition of sufficient statistics.

Example 2.4 (Classical sufficient statistics as one-step Bellman states). Consider a nonadaptive dominated experiment with observations $X_1, \dots, X_n$ from P_θ , and no control. Let $T_n = T(X_1, \dots, X_n)$ satisfy the Fisher–Neyman factorization. If the loss and index depend on θ only through quantities whose posterior distribution is determined by T_n , then the process $S_t = (t, T_{t-1})$ is Bellman sufficient. Predictive sufficiency is the usual conditional factorization; update sufficiency is $T_t = u(T_{t-1}, X_t)$; posterior predictive and index sufficiency follow because the posterior or conditional reference law depends on the history only through T_t . In a regular exponential family,

$$p_\theta(x) = h(x) \exp\{\vartheta(\theta)^\top T(x) - A(\theta)\},$$

with conjugate or otherwise statistic-measurable reference law, the cumulative statistic $\sum_{i < t} T(X_i)$ gives the corresponding state. This is the static prototype for all examples below.

Example 2.5 (Finite stochastic multi-armed bandits). Let $\Omega \subseteq \Theta^K$ and let arm a produce an observation from P_{θ_a} . For unit-variance Gaussian arms, $P_{\theta_a} = N(\theta_a, 1)$. A reference Gaussian prior with independent coordinates gives the posterior state

$$S_t = (t, (n_{t,a}, \bar{X}_{t,a}, v_{t,a})_{a=1}^K),$$

where $n_{t,a}$ is the number of previous pulls of arm a , $\bar{X}_{t,a}$ is the empirical mean when $n_{t,a} > 0$, and $v_{t,a}$ is the posterior variance. Equivalently one may store the posterior hyperparameters. For a fixed truth $\omega = \theta$,

$$P_{\omega,s,a} = N(\theta_a, 1), \quad \ell_\omega(s, a) = \max_b \theta_b - \theta_a,$$

and the update of $(n, \bar{X}, v)$ after observing arm a is a function of (s, a, o) . The posterior predictive law is $N(m_{t,a}, 1 + v_{t,a})$, and for the action index $Y = A^*(\theta)$ the conditional predictive law $P_{s,a}^y$ and conditional loss $\ell_y^X(s, a)$ are obtained by integrating the same Gaussian posterior over the event $A^*(\theta) = y$. Hence Definition 2.3 is satisfied. The empirical mean alone is not sufficient: the count or posterior variance is needed for prediction, confidence, and information gain.

Example 2.6 (Finite-dimensional Gaussian linear bandits). Let $\Omega \subseteq \mathbb{R}^d$, actions satisfy $\|a\|_2 \leq 1$, and

$$O_t = \langle \theta, A_t \rangle + \xi_t, \quad \xi_t \sim N(0, 1).$$

Under a Gaussian reference prior, the exact Bayesian state is $S_t = (t, m_t, \Sigma_t)$, where

$$\Sigma_{t+1}^{-1} = \Sigma_t^{-1} + A_t A_t^\top, \quad m_{t+1} = \Sigma_{t+1}^{-1} \{ \Sigma_t^{-1} m_t + A_t O_t \}.$$

For fixed θ , $P_{\theta,s,a} = N(\langle \theta, a \rangle, 1)$ and the regret loss is $\sup_{\|b\|_2 \leq 1} \langle \theta, b \rangle - \langle \theta, a \rangle$. The posterior predictive law is $N(\langle m_t, a \rangle, 1 + a^\top \Sigma_t a)$, and conditioning the Gaussian posterior on an index event such as $\chi(\theta) = \theta$, $\chi(\theta) = \theta/\|\theta\|_2$, or $\chi(\theta) = A^*(\theta)$ determines $P_{s,a}^y$ and $\ell_y^x(s, a)$. Therefore the Gaussian mean-covariance pair is an exact Bellman-sufficient state. A frequentist UCB state $(\hat{\theta}_t, V_t, \beta_t)$ is different: it can certify a fixed-truth bracket on the calibration event $\|\hat{\theta}_t - \theta^*\|_{V_t} \leq \beta_t$, but it is not an exact posterior-reference state unless a reference belief or confidence-to-belief map is added.

Example 2.7 (Finite active marginals in kernel bandits). Let f^* belong to an RKHS on a possibly infinite domain, but suppose the learner is evaluated on a finite active set $\mathcal{X} = \{x_1, \dots, x_n\}$. Under the GP reference law on the active reward vector $F = (f(x_1), \dots, f(x_n))$, the finite marginal posterior $(m_t, \Sigma_t) \in \mathbb{R}^n \times \mathbb{R}^{n \times n}$ is Bellman sufficient for the finite experiment. For fixed f^* , pulling x_i has law $N(f^*(x_i), \lambda)$ in the Gaussian reference calculation; the posterior predictive law is $N(m_{t,i}, \lambda + \Sigma_{t,ii})$; and the usual rank-one Gaussian update is a function of (m_t, Σ_t, i, O_t) . If the index is the optimal active action $Y = \arg \max_i F_i$, then index sufficiency follows by conditioning the finite Gaussian posterior on $\arg \max_i F_i = y$. The infinite-dimensional RKHS function is still the frequentist truth; the sufficient state is finite only because the decisions and observations use the active marginal.

2.5 Full environment posterior is fallback sufficient state

For sequential decision making, a natural Bellman-sufficient representation is obtained by retaining the full environment posterior. For suitably defined Markovian model classes over the state space, which include standard episodic model-based reinforcement learning settings, this posterior serves as a fallback sufficient state.

Example 2.8 (Full environment posterior as a fallback sufficient state). *Consider a realizable model class in which each environment $\omega \in \Omega$ specifies, for every retained physical state or context z , action a , and stage t , an observation kernel $P_{\omega,t}(\cdot | z, a)$, a loss $\ell_{\omega,t}(z, a)$, and the physical-state transition rule. Let Z_t denote the physical state, context, stage, or episode coordinate needed to make these kernels Markov. Fix a reference prior μ and set*

$$\Pi_t = \mathcal{L}_\mu(\Omega | H_{t-1}), \quad S_t = (Z_t, \Pi_t).$$

Assume the observation laws are dominated on the relevant support. Then S_t gives an exact Bellman-sufficient state. Indeed, for each fixed truth ω ,

$$P_{\omega, S_t, a} = P_{\omega, t}(\cdot | Z_t, a), \quad \ell_\omega(S_t, a) = \ell_{\omega, t}(Z_t, a),$$

and the next retained state is obtained by updating the physical coordinate and the posterior

$$\Pi_{t+1}(d\omega) = \frac{p_{\omega, t}(O_t | Z_t, A_t) \Pi_t(d\omega)}{\int p_{\omega', t}(O_t | Z_t, A_t) \Pi_t(d\omega')}.$$

For an index $\chi : \Omega \rightarrow \mathcal{Y}$, the indexed marginal and conditional predictive components are

$$q_t = \chi_{\#} \Pi_t, \quad \Pi_t^y = \mathcal{L}_{\Pi_t}(\Omega \mid \chi(\Omega) = y),$$

$$P_{S_t, a}^y = \int P_{\omega, t}(\cdot \mid Z_t, a) \Pi_t^y(d\omega), \quad \ell_y^x(S_t, a) = \int \ell_{\omega, t}(Z_t, a) \Pi_t^y(d\omega).$$

Thus prediction, loss evaluation, posterior/reference updating, and indexed information accounting all close on S_t . This verifies Definition 2.3.

In this sense, sequential decision making with Bellman-sufficient representations contains the Decision Making with Structured Observations (DMSO) framework (Foster et al., 2021) as a special case when DMSO is interpreted as an independent episodic model-learning formalism and the full environment posterior is retained as the state. At the same time, the present framework also accommodates Markovian dependence across physical states and inner horizon stages, as in computational dynamic programming. The more important point is not formal inclusion, but the possibility of running dynamic programming on sufficient states and of charging a decision-relevant index rather than the full model label.

Example 2.9 (DMSO as a posterior-state specialization). *In the independent-episode DMSO formalism (Foster et al., 2021), an environment M specifies the observation law and loss for each decision rule within an episode, and episodes are conditionally independent given M . Under the observation-packet convention, the episode history H_{k-1} already contains the public information available before episode k . For a fixed prior and likelihood, the posterior*

$$\Pi_k = \mathcal{L}(M \mid H_{k-1})$$

is a deterministic compression of that history, not additional oracle information. Taking Ω to be the DMSO model class and retaining Π_k , together with any public episode context or admissible-decision constraints, gives a literal Bellman-sufficient state for the reference experiment. The posterior predictive observation law, the posterior update after the episode, and every index marginal $\chi_{\#} \Pi_k$ are functions of this state. Thus DMSO fits the present framework as the special case in which the Bellman clock is the outer episode and the posterior is the retained state coordinate. More compressed representations require an additional sufficiency proof; they do not follow from the DMSO notation alone.

Since the general one-step primitives in (1) allow the environment, or an adaptive adversary, to depend on both the history H_{k-1} and the announced mixed action p_k as a current control, the adversarial DMSO setting (Foster et al., 2022; Xu and Zeevi, 2025) is also subsumed; see Remark 2.1. In this case, one may take the full environment to be the entire nonanticipating model sequence, with the posterior over this environment serving as a valid posterior state. The adversarial DMSO framework is still episodic rather than across-episode dynamic. It captures best-in-hindsight comparison in the classical online-learning sense, but does not by itself model planning problems in which the comparator is dynamic and current decisions shape future states across episodes. Extending the framework to such dynamic planning settings would require further development along the lines pursued in this paper.

2.6 Why sufficient states and compression matter

The full history H_{t-1} is always a valid Bellman state, so state compression is not needed for the formal existence of a dynamic program. The theory developed in this paper could also be

stated using the full history H_{t-1} together with full-history information indices. Indeed, any state-measurable information index is automatically an index on the full history: if $S_t = \phi_t(H_{t-1})$, then a state-dependent index $\chi_{S_t}(\Omega)$ lifts to the full-history index $\chi_{\phi_t(H_{t-1})}(\Omega)$. Conversely, a full-history index descends to the compressed state only if it is constant on the fibers of the compression. The role of sufficient states is therefore representational rather than existential: they identify a level of compression at which the relevant Bellman recursions and information accounting close. Such a state need not be unique, nor must it be minimal.

It is already clear that sufficient states are useful at a practical level: specifying them makes it tractable to identify the relevant information index, construct upper-bound algorithms with clear links to popular design principles, and formulate lower-bound methods. The question in this subsection is more conceptual: whether sufficient states offer any information-theoretic advantage beyond the full history. The answer is yes. By data processing, compression can only reduce the information budget; when the retained state is Bellman-sufficient in the appropriate controlled sense, this reduction need not lose any of the Bellman structure required for upper or lower bounds. The key is to distinguish Bellman sufficiency from classical conditional-independence sufficiency. Otherwise, if the relevant information gain were always invariant under compression, sufficient states would be more a methodological convenience than an intrinsic representation-level object. A general formal claim that every proof device operating on the full history, but not passing through the constructed sufficient state, must lead to coarser bounds would be too strong. We do not pursue such a claim here.

It is helpful to separate three notions. Classical model sufficiency is the static conditional-independence condition

$$\Omega \perp H_{t-1} \mid S_t,$$

which says that the state retains all posterior information about the full environment. Index-marginal sufficiency for $Y = \chi(\Omega)$ is weaker:

$$Y \perp H_{t-1} \mid S_t.$$

It says only that the state retains the full-history posterior marginal of the index. Bellman sufficiency, as used in Definition 2.3, is different: it is the dynamic closure of the primitives needed for decision making, namely feasible actions, fixed-truth predictive kernels, losses or surrogate losses, updates, and, when needed, the indexed reference objects used by the logarithmic potential and the Bellman–Fano method. Thus a Bellman-sufficient state need not be sufficient for the full model, and a state-based reference process may be coarser than the full-history posterior; its validity comes from the Bellman inequalities that close on that state.

Proposition 2.10 (Bellman quotient and information monotonicity). *Let $S_t = \phi_t(H_{t-1})$ be a fixed-truth Bellman-sufficient representation. Suppose a one-step Bellman cost c_ω and a continuation value F_{t+1} are state-measurable, so that*

$$c_\omega(h, p) = c_\omega(s, p), \quad F_{t+1}(H_t) = f_{t+1}(S_{t+1}), \quad s = S_t(h).$$

Then the full-history Bellman operator factors through S_t : for every h with $S_t(h) = s$,

$$\inf_{p \in \Delta(\mathcal{A}_t(h))} \sup_{\omega \in \mathcal{C}_t(h)} \{c_\omega(h, p) + \mathbb{E}_{\omega, h, p} F_{t+1}(H_t)\} = \inf_{p \in \Delta(\mathcal{A}_t(s))} \sup_{\omega \in \mathcal{C}_t(s)} \{c_\omega(s, p) + \mathbb{E}_{\omega, s, p} f_{t+1}(S_{t+1})\},$$

whenever the comparison set is also state-measurable. Moreover, for any reference law and any fixed index $Y = \chi(\Omega)$,

$$I(Y; S_t) \leq I(Y; H_{t-1}).$$

Equality in the information display holds precisely under index-marginal sufficiency, $Y \perp H_{t-1} \mid S_t$.

Proof. The Bellman-operator identity follows directly from the state-measurability of the action set, comparison set, one-step cost, transition law, and continuation value. Histories in the same fiber of S_t induce the same local optimization problem. The information inequality is the data-processing inequality applied to the measurable map $S_t = \phi_t(H_{t-1})$. The equality condition is the standard equality case for conditional mutual information, $I(Y; H_{t-1} \mid S_t) = 0$. $\square$

This proposition is the formal reason sufficient states are useful. The quotient preserves the Bellman problem whenever the primitives close on the state, while data processing ensures that the state-based information accounting for a fixed index is no larger than the full-history accounting. If exact index-marginal sufficiency holds, the two information quantities are equal; if the state intentionally uses a coarser algorithmic or reference belief, the information quantity may be strictly smaller. That strict reduction is not a free theorem about the original unrestricted problem: it is valid only when the upper Bellman recursion and the lower Bellman–Fano recursion are both proved on the same coarsened state. In this sense the framework is representational. One searches for a state and index on which the dynamic program and the entropy accounting simultaneously close; when they do, the resulting sandwich can be tighter and more intrinsic than a certificate written directly on the full history, even for the same global index Y .

3 Information complexity sandwich

This section isolates the comparison used throughout the paper. After a Bellman-sufficient state, an index, a reference update, and a calibration mechanism have been fixed, the upper and lower bounds are expressed in the same units. The upper logarithmic Bellman theorem pays an initial coordinate code length for the realized index. The lower Bellman–Fano theorem is governed by the ghost entropy of reference histories that are already good for the sampled index. The comparison is therefore an information-risk sandwich: an upper Bellman value at an index-code budget is compared with a lower Bellman–Fano value at a ghost-entropy budget. This section makes the finite-index logarithmic gap, the constant-gap localization condition, and the growth/fixed-point requirements explicit.

3.1 Code length, hard priors, and ghost entropy

Let $\Omega_0 \subseteq \Omega$ be the problem class being lower and upper bounded, let $\chi : \Omega \rightarrow \mathcal{Y}$ be the index, and let q_1 be an initial reference marginal on $\mathcal{Y}$. The fixed-truth upper telescope pays the worst-case initial coordinate code length

$$L_0(\Omega_0; q_1, \chi) := \sup_{\omega \in \Omega_0} \log \frac{1}{q_1(\chi(\omega))}, \quad (7)$$

with the convention that the value is infinite if $q_1(\chi(\omega)) = 0$ for some $\omega \in \Omega_0$.

For a hard prior μ supported on Ω_0 , write $q_1 = \chi_{\#}\mu$. At an average-loss radius r , define the ghost-good probability and ghost entropy by

$$p_{\mu,r} := \mathbb{P}_{Y \sim q_1, H'_T} \{\bar{L}_\chi(Y, H'_T) \leq r\}, \quad E_{\mu,r} := \log \frac{1}{p_{\mu,r}}. \quad (8)$$

Here H'_T is the reference history used in the Bellman–Fano method and is independent of Y under the ghost law. For an algorithm-uniform theorem, $p_{\mu,r}$ may be replaced by the exact Bellman ghost value $\Gamma_1^{r,*}(q_1, s_1, 0)$, or by any valid upper certificate $\bar{\Gamma}_1^r(q_1, s_1, 0) \geq \Gamma_1^{r,*}(q_1, s_1, 0)$. The corresponding certified entropy is $-\log \bar{\Gamma}_1^r$.

Definition 3.1 (Bellman–Fano admissible hard prior). *Fix $r > 0$. A prior μ supported on Ω_0 is Bellman–Fano admissible at radius r if the lower Bellman recursions, or valid supersolutions of them, provide a nonnegative indexed-information certificate C_μ and a ghost-entropy certificate $E_{\mu,r}$ such that*

$$C_\mu + \log 2 \leq \frac{1}{2} E_{\mu,r}. \quad (9)$$

The intrinsic choice is $C_\mu = C_1^(s_1)$ and $E_{\mu,r} = -\log \Gamma_1^{r,*}(q_1, s_1, 0)$, where C^* and $\Gamma^{r,*}$ are the exact information-capacity and ghost-good Bellman values from Section 6.3. Computable proofs may use conservative choices $C_\mu = \bar{C}_1(s_1)$ and $E_{\mu,r} = -\log \bar{\Gamma}_1^r(q_1, s_1, 0)$.*

The condition $p_{\mu,r} \leq \varrho < 1$ sometimes appears as a convenient nondegeneracy diagnostic: it prevents the lower entropy denominator from vanishing. It plays the same mathematical role as the nondegeneracy requirements in offset or constrained DEC comparisons (Foster et al., 2021, 2023): without positive decision separation at the chosen information scale, the denominator of the rate comparison is zero and no lower bound can be extracted. In the Bellman–Fano formulation this condition need not be imposed separately once the hard prior is defined through admissibility.

Lemma 3.2 (Admissibility implies nondegenerate ghost entropy). *If μ is Bellman–Fano admissible at radius r , then*

$$E_{\mu,r} \geq 2 \log 2, \quad p_{\mu,r} \leq \frac{1}{4}.$$

Consequently a separate assumption $p_{\mu,r} \leq \varrho < 1$ is unnecessary for any admissible hard prior. Conversely, if one starts from a proposed prior and only knows $p_{\mu,r} \leq \varrho < 1$, then the entropy denominator is at least $\log(1/\varrho)$, but this by itself does not prove Bellman–Fano admissibility because the information certificate C_μ must also satisfy (9).

Proof. Since $C_\mu \geq 0$, condition (9) implies $E_{\mu,r} \geq 2 \log 2$. The relation $p_{\mu,r} = e^{-E_{\mu,r}}$ gives $p_{\mu,r} \leq e^{-2 \log 2} = 1/4$. The converse statement is only the definition of the entropy lower bound $E_{\mu,r} \geq \log(1/\varrho)$; it does not control C_μ . $\square$

Proposition 3.3 (Finite-index entropy comparison with an explicit hard prior). *Let $\mathcal{Y}_h \subseteq \chi(\Omega_0)$ be finite with $|\mathcal{Y}_h| = M \geq 2$. Choose one representative environment $\omega^y \in \Omega_0$ for each $y \in \mathcal{Y}_h$ such that $\chi(\omega^y) = y$, and let μ_h be the uniform prior on $\{\omega^y : y \in \mathcal{Y}_h\}$. Then q_1 is uniform on $\mathcal{Y}_h$, and*

$$L_0(\{\omega^y : y \in \mathcal{Y}_h\}; q_1, \chi) = \log M.$$

If μ_h is Bellman–Fano admissible at radius r , then the entropy mismatch between the upper coordinate code length and the lower ghost entropy satisfies

$$\frac{L_0}{E_{\mu_h,r}} \leq \frac{\log M}{2 \log 2}. \quad (10)$$

Thus, without pursuing constant matching, admissibility alone gives an at-most-logarithmic finite-index entropy gap. More sharply, if for every reference history h' ,

$$G_r(h') := \{y \in \mathcal{Y}_h : \bar{L}_\chi(y, h') \leq r\}$$

has cardinality at most $m_r < M$, then

$$E_{\mu_h,r} \geq \log \frac{M}{m_r}, \quad \frac{L_0}{E_{\mu_h,r}} \leq \frac{\log M}{\log(M/m_r)}. \quad (11)$$

In particular, constant-factor entropy matching requires $E_{\mu_h,r} \asymp \log M$, for example through $m_r \leq M^{1-\alpha}$ for some constant $\alpha > 0$. A mere nondegeneracy bound $p_{\mu_h,r} \leq \varrho < 1$ gives only $L_0/E_{\mu_h,r} \leq \log M / \log(1/\varrho)$.

Proof. The uniform representative prior gives $q_1(y) = 1/M$, hence the upper code length is $\log M$. If μ_h is Bellman–Fano admissible, Lemma 3.2 gives $E_{\mu_h, r} \geq 2 \log 2$, proving (10). Under the multiplicity condition,

$$p_{\mu_h, r} = \mathbb{E}_{H'_T} \frac{|G_r(H'_T)|}{M} \leq \frac{m_r}{M}.$$

Taking $-\log$ gives $E_{\mu_h, r} \geq \log(M/m_r)$, which proves (11). The final statement follows from $E_{\mu_h, r} \geq \log(1/\varrho)$. $\square$

3.2 From entropy comparison to regret comparison

An entropy comparison alone is not a regret comparison. The missing ingredient is a regularity property of the upper Bellman value as its information budget changes. This is the same structural reason that constrained DEC bounds are stated through a coefficient, localization, or fixed-point condition rather than through a bare logarithmic-model-cardinality entropy term (Foster et al., 2023): cardinality controls the amount of information charged, but a risk modulus is still needed to turn that information budget into a regret radius.

For a fixed state/index representation and the log-penalized Bellman program in Section 5.1, write

$$\mathsf{U}_T(L) := \inf_{\gamma > 0} \{W_1^\gamma(s_1) + \gamma L + \Delta_T^\gamma\}, \quad (12)$$

where W^γ is the log-penalized Bellman value and $\Delta_T^\gamma \geq 0$ collects worst-case calibration, approximation, and optimization errors for that value as in Theorem 5.2. For the exact calibrated Bellman dynamic program, $\Delta_T^\gamma = 0$. We assume U_T is nondecreasing in L , as holds for the optimized upper value in the applications considered here.

Assumption 3.4 (Bellman upper-growth condition). *There exist constants $C_{\text{gr}} \geq 1$ and $\beta \in [0, 1]$ such that, for all $L > 0$ and $a \geq 1$,*

$$\mathsf{U}_T(aL) \leq C_{\text{gr}} a^\beta \mathsf{U}_T(L).$$

This is a regularity condition on the chosen upper value. It is not a consequence of finite index cardinality alone; it is the Bellman analogue of the sub-root or growth condition used to convert entropy or localized information radii into risk radii in fixed experiments and in constrained DEC analyses.

Lemma 3.5 (Why a growth condition is needed). *No regret-ratio comparison follows from an entropy-ratio comparison for an arbitrary nondecreasing upper value U_T . More precisely, fix $0 < E < L$ and any $B > 0$. There is a nondecreasing function U such that $\mathsf{U}(L)/\mathsf{U}(E) \geq B$.*

Proof. Set $\mathsf{U}(x) = 1$ for $x < L$ and $\mathsf{U}(x) = B$ for $x \geq L$. Then U is nondecreasing, $\mathsf{U}(E) = 1$, and $\mathsf{U}(L) = B$. Thus a logarithmic entropy gap, even $L/E = O(\log M)$, does not by itself imply any controlled regret gap. Assumption 3.4, or an equivalent fixed-point/rate condition, is the additional structure that rules out such jumps. $\square$

Definition 3.6 (Upper-at-lower-entropy calibration). *Let μ be Bellman–Fano admissible at radius r with ghost entropy $E_{\mu, r}$. The upper value is calibrated at the lower entropy scale if*

$$\mathsf{U}_T(E_{\mu, r}) \leq C_{\text{base}} Tr \quad (13)$$

for a universal or problem-controlled constant C_{base} . Equivalently, the phrase “the upper and lower values close at the same radius” means (13); it is not an additional philosophical assumption. It can be verified directly, by a localized or constrained information-gain estimate, or by choosing r as a fixed point of the rate equation $\mathsf{U}_T(E_{\mu, r}) \asymp Tr$.

Proposition 3.7 (Three equivalent scale-closing diagnostics). *For a Bellman–Fano admissible pair (μ, r) , each of the following conditions implies upper-at-lower-entropy calibration (13).*

- (i) *Direct budget control:* $U_T(E_{\mu,r}) \leq C_{\text{base}} Tr$.
- (ii) *Coefficient control:* *there exists $\gamma > 0$ such that*

$$W_1^\gamma(s_1) + \gamma E_{\mu,r} + \Delta_T^\gamma \leq C_{\text{base}} Tr.$$

- (iii) *Fixed-point control:* *r is chosen so that $U_T(E_{\mu,r})/T \leq C_{\text{base}} r$.*

Condition (ii) is the form closest to constrained DEC and localized information-gain analyses: the same localized comparison that certifies the information budget also bounds the decision term at radius r . Condition (iii) packages the same requirement as a rate fixed point.

Proof. Condition (i) is exactly (13). Condition (ii) implies (i) because $U_T(E_{\mu,r})$ is the infimum over $\gamma > 0$ of the left side in (12). Condition (iii) is a restatement of (i) with the constant displayed after dividing by T . $\square$

Theorem 3.8 (Frequentist Bellman information-risk sandwich). *Fix $\Omega_0 \subseteq \Omega$, a horizon T , an index χ , a Bellman-sufficient state/reference update, and an initial index law q_1 with finite $L_0 = L_0(\Omega_0; q_1, \chi)$. Suppose the upper side admits a calibrated log-penalized Bellman certificate whose budget value is U_T in (12). Then*

$$\mathfrak{R}_T^*(\Omega_0) \leq U_T(L_0). \quad (14)$$

Conversely, if a prior μ supported on Ω_0 is Bellman–Fano admissible at radius r , then

$$\mathfrak{R}_T^*(\Omega_0) \geq \frac{Tr}{2}. \quad (15)$$

If, in addition, Assumption 3.4 holds and the upper value is calibrated at the lower entropy scale in the sense of Definition 3.6, then

$$\mathfrak{R}_T^*(\Omega_0) \leq C_{\text{gr}} C_{\text{base}} \left(\max \left\{ 1, \frac{L_0}{E_{\mu,r}} \right\} \right)^\beta Tr, \quad (16)$$

and hence the ratio between the displayed upper value and the Bellman–Fano lower bound $Tr/2$ is at most

$$2C_{\text{gr}} C_{\text{base}} \left(\max \left\{ 1, \frac{L_0}{E_{\mu,r}} \right\} \right)^\beta.$$

For a finite uniform hard index prior of size M , admissibility alone gives the crude logarithmic comparison

$$\frac{L_0}{E_{\mu,r}} \leq \frac{\log M}{2 \log 2},$$

while the multiplicity/localization condition $|G_r(h')| \leq m_r$ gives

$$\frac{L_0}{E_{\mu,r}} \leq \frac{\log M}{\log(M/m_r)}.$$

Thus logarithmic-gap matching requires only Bellman–Fano admissibility plus upper growth and upper-at-lower-entropy calibration. Constant-factor matching requires the stronger localization $E_{\mu,r} \asymp \log M$, or an equivalent constraint/fixed-point argument that certifies the same entropy order.

Proof. The upper inequality is Theorem 5.2, optimized over γ , with the uniform coordinate code length bounded by L_0 and the remaining errors included in Δ_T^γ . The lower inequality is Theorem 6.3 applied to the Bellman–Fano admissible prior, followed by Yao’s principle because μ is supported on Ω_0 . For the final comparison, set

$$a = \max \left\{ 1, \frac{L_0}{E_{\mu,r}} \right\}.$$

Since $\mathcal{U}_T$ is nondecreasing and $aE_{\mu,r} \geq L_0$, Assumption 3.4 and (13) give

$$\mathcal{U}_T(L_0) \leq \mathcal{U}_T(aE_{\mu,r}) \leq C_{\text{gr}} a^\beta \mathcal{U}_T(E_{\mu,r}) \leq C_{\text{gr}} C_{\text{base}} a^\beta Tr.$$

Combining this with $\mathfrak{R}_T^*(\Omega_0) \geq Tr/2$ gives the regret-ratio bound. The finite-index statements are Proposition 3.3. $\square$

Relation to offset and constrained DEC. The condition $p_{\mu,r} \leq \varrho < 1$ is not a separate structural assumption in the final sandwich; it is a quick diagnostic that a proposed hard prior has nonzero ghost entropy. Bellman–Fano admissibility is the cleaner replacement because it simultaneously checks nonzero ghost entropy and small indexed information. This is analogous to offset or constrained DEC theory in the following nondegeneracy and localization sense: a rate comparison is meaningful only after the decision-separation term at the chosen scale is nondegenerate and the information term is localized or constrained at the same scale (Foster et al., 2021, 2023).

Constrained DEC incorporates the nondegeneracy and localization checks into the coefficient or fixed-point definition, while the accompanying upper-growth condition controls the information-to-regret conversion (Foster et al., 2023). For a finite model-index class, this can still leave a worst-case logarithmic gap in the upper–lower sandwich, measured in the model cardinality. A tight theory of DEC is technically challenging partly because it seeks a one-step and state-uniform coefficient that simultaneously captures decision loss, statistical indistinguishability, localization, and adaptive information accumulation. This is a stronger requirement than assigning a single global dimension to the model class. Even in classical statistical learning, sharp rates are often governed by localized, distribution-dependent, data-dependent, or metric-dependent quantities, such as empirical or local Rademacher complexities and chaining functionals for the relevant induced metric. Thus a worst-case cardinality or global-dimension gap under regularity conditions is a rigorous universal fallback, but it can be too coarse for precise applications. This is one motivation for replacing a single global coefficient by a dynamic, state-dependent Bellman information certificate.

The Bellman sandwich makes explicit three checks that coefficient-based theories often package together: (i) hard-prior admissibility on the lower side, which combines information control with nonvacuous ghost entropy and localization; (ii) upper-at-lower-entropy calibration, which checks that the log-penalized Bellman upper value is of the right order at the entropy scale certified by the hard prior; and (iii) upper growth, which converts an entropy mismatch into a regret mismatch. The separation is useful because the lower side may be proved by Bellman–Fano ghost entropy, while the upper side may be proved by exact log-potential Bellman programming, UCB, E2D/DEC, AMS/EBO, or another relaxation. DEC theory removes the need for a separate upper-at-lower-entropy calibration condition: the lower bound is derived from the DEC indistinguishability game, while the matching upper bound is attained by an E2D-type algorithm controlled by the same coefficient. However, for finite model-index classes, the resulting analysis can still retain a logarithmic dependence on the model cardinality. Thus the finite-cardinality conclusion of this section is conditional, but clean: under a uniform hard index prior and Bellman–Fano admissibility, the entropy gap is at most logarithmic in the index cardinality. With localization strong enough to give $E_{\mu,r} \asymp \log M$, this gap becomes constant, as illustrated by the basic examples in Section 7.

4 Indexed information and exact AIR/MAIR identities

4.1 Posterior-reference histories and index information

The mutual-information identities in this subsection use the exact posterior-reference version of Definition 2.3. The fixed-truth log-gain regret identity in the next subsection should not be substituted into the posterior chain rule unless the two objects are induced by a coherent reference law.

For a prior μ and algorithm Alg, define the Bayesian posterior-reference trajectory law

$$\bar{\mathbb{P}}_\mu^{\text{Alg}} := \int \mathbb{P}_\omega^{\text{Alg}} \mu(d\omega).$$

A reference history is denoted

$$H'_T \sim \bar{\mathbb{P}}_\mu^{\text{Alg}}, \quad S'_t = \phi_t(H'_{t-1}, b'_t), \quad b'_t = \mathcal{L}_\mu(\Omega \mid H'_{t-1}),$$

and the algorithm's reference decision kernel is

$$p'_t = p_t(\cdot \mid H'_{t-1}).$$

Definition 4.1 (Indexed information gain). *At a current time-state (t, s) , let q_s be the posterior law of $Y = \chi(\Omega)$ and let $P_{s,a}^y$ and $P_{s,a}$ be the conditional and marginal predictive laws from Definition 2.3; we suppress t in these laws when the stage is part of s . For $p \in \Delta(\mathcal{A}_t(s))$, define*

$$\mathcal{I}_\chi(s, p) := \mathbb{E}_{a \sim p} \int D_{\text{KL}}(P_{s,a}^y \parallel P_{s,a}) q_s(dy). \quad (17)$$

When $Y = \Omega$, this is model-index information. When $Y = \pi^*(\Omega)$, this is action-index information. In the action-index case, the belief is still a belief over environments; only the information target is the optimal action.

Define the cumulative and average indexed information of an algorithm by

$$C_\chi^{\text{Alg}}(\mu) := \mathbb{E}_{H'_T \sim \bar{\mathbb{P}}_\mu^{\text{Alg}}} \sum_{t=1}^T \mathcal{I}_\chi(S'_t, p'_t), \quad \bar{C}_\chi^{\text{Alg}}(\mu) := \frac{1}{T} C_\chi^{\text{Alg}}(\mu). \quad (18)$$

Proposition 4.2 (Reference-history indexed chain rule). *Assume that the retained state is an exact posterior-indexed lift in the sense of Definition 2.3, and that the displayed KL terms are well defined and integrable. For every prior μ , index map χ , and algorithm Alg,*

$$I_\mu(Y; H_T) = \mathbb{E}_{H'_T \sim \bar{\mathbb{P}}_\mu^{\text{Alg}}} \sum_{t=1}^T \mathcal{I}_\chi(S'_t, p'_t) = T \bar{C}_\chi^{\text{Alg}}(\mu). \quad (19)$$

Proof. The mutual-information chain rule gives

$$I_\mu(Y; H_T) = \sum_{t=1}^T I_\mu(Y; A_t, O_t \mid H_{t-1}).$$

Given H_{t-1} , the action A_t is sampled by the algorithm using only randomization independent of the environment and is conditionally independent of Y . Therefore

$$I_\mu(Y; A_t, O_t \mid H_{t-1}) = I_\mu(Y; O_t \mid H_{t-1}, A_t).$$

Conditioning further on $A_t = a$, the law of O_t given $Y = y$ is $P_{S_t,a}^y$, while the posterior predictive law is $P_{S_t,a}$. Hence

$$I_\mu(Y; O_t \mid H_{t-1}, A_t = a) = \int D_{\text{KL}}(P_{S_t,a}^y \parallel P_{S_t,a}) q_{S_t}(dy).$$

Averaging over $a \sim p_t(\cdot \mid H_{t-1})$ and over the marginal law of the history under the Bayesian mixture law proves the result. Writing this marginal law as the reference history law gives the displayed expression. $\square$

4.2 Fixed-truth indexed AIR bracket

The posterior-averaged information gain in (17) is the correct object for Bayesian chain rules and Bellman–Fano lower-bound method. The upper Bellman program uses the corresponding fixed-truth coordinate identity. The coordinate identity is most cleanly stated in AIR form: a current reference marginal on the index is scored by the logarithmic posterior of the index after the current action and observation.

Coefficient convention. Throughout the upper-bound identities and Bellman programs, the information multiplier is denoted by $\gamma > 0$: a one-step log gain G is charged as γG , and an initial coordinate code length is charged as $\gamma \log(1/q)$. Some AIR/MAIR/DEC papers use a temperature parameter η with multiplier $1/\eta$ (Xu and Zeevi, 2025; Liu et al., 2025, 2026); in the present notation this is simply $\gamma = 1/\eta$.

Fix a state s , an action distribution $p \in \Delta(\mathcal{A}_t(s))$, and a pair belief

$$\nu \in \Delta(\Omega \times \mathcal{Y})$$

over environments and index values. Let q_ν be the $\mathcal{Y}$ -marginal of ν , and let $q \in \text{int } \Delta(\mathcal{Y})$ be the current reference marginal used by the logarithmic score. The pair belief induces, for each realized action a , the predictive mixture

$$P_{\nu,s,p,a}(\cdot) := \int P_{\omega,t}(\cdot \mid s, p, a) \nu(d\omega, dy),$$

and the posterior index marginal

$$q_\nu^+(B \mid s, p, a, o) := \nu(Y \in B \mid s, p, a, O = o), \quad B \subseteq \mathcal{Y}.$$

When the observation laws are dominated by a common measure and have densities $p_{\omega,t,s,p,a}(o)$, this update is explicitly

$$q_\nu^+(B \mid s, p, a, o) = \frac{\int_{\Omega \times B} p_{\omega,t,s,p,a}(o) \nu(d\omega, dy)}{\int_{\Omega \times \mathcal{Y}} p_{\omega,t,s,p,a}(o) \nu(d\omega, dy)}. \quad (20)$$

In ordinary stochastic bandits and MDPs the dependence on p is suppressed.

Let $\ell_{\omega,y}(s, p)$ denote the comparison loss associated with pair (ω, y) ; for a valid fixed truth one takes $y = \chi(\omega)$ and $\ell_{\omega,\chi(\omega)} = \ell_\omega$. Define the general indexed AIR functional

$$\mathfrak{A}_{q,\gamma}(s, p, \nu) := \mathbb{E}_{(\Omega,Y) \sim \nu} \ell_{\Omega,Y}(s, p) - \gamma \mathbb{E}_{(\Omega,Y) \sim \nu, a \sim p, O \sim P_{\Omega,t}(\cdot \mid s, p, a)} \log \frac{q_\nu^+(Y \mid s, p, a, O)}{q(Y)}. \quad (21)$$

Equivalently,

$$\mathfrak{A}_{q,\gamma}(s, p, \nu) = \mathbb{E}_\nu \ell_{\Omega, Y}(s, p) - \gamma \mathcal{I}_\nu(Y; O \mid s, p) - \gamma D_{\text{KL}}(q_\nu \| q), \quad (22)$$

where

$$\mathcal{I}_\nu(Y; O \mid s, p) := \mathbb{E}_{a \sim p} \int D_{\text{KL}}(P_{\nu, s, p, a}^y \| P_{\nu, s, p, a}) q_\nu(dy)$$

is the one-step posterior-averaged information computed under the candidate pair belief. Thus q is the current reference index marginal, while q_ν^+ is constructed from the algorithmic pair belief ν and the current likelihood model. In an exact Bayesian specialization, ν is the current posterior over $(\Omega, \chi(\Omega))$ and $q = q_\nu$. In AIR/AMS/EBO, ν may be an optimized or robust candidate pair belief, and the KL term in (22) accounts for moving its index marginal away from the current reference q .

Lemma 4.3 (AIR gradient bracket). *Assume finite spaces, or dominated spaces where the displayed Gateaux derivatives exist. Up to an additive constant on the probability simplex,*

$$\frac{\partial}{\partial \nu(\omega, y)} \mathfrak{A}_{q,\gamma}(s, p, \nu) = \ell_{\omega, y}(s, p) - \gamma \mathbb{E}_{a \sim p, O \sim P_{\omega, t}(\cdot | s, p, a)} \log \frac{q_\nu^+(y \mid s, p, a, O)}{q(y)}. \quad (23)$$

Consequently, for every fixed pair (ω, y) ,

$$\begin{aligned} B_{q,\gamma}^{\text{AIR}}(s, p, \nu; \omega, y) &:= \mathfrak{A}_{q,\gamma}(s, p, \nu) + \langle \nabla_\nu \mathfrak{A}_{q,\gamma}(s, p, \nu), \delta_{(\omega, y)} - \nu \rangle \\ &= \ell_{\omega, y}(s, p) - \gamma \mathbb{E}_{a \sim p, O \sim P_{\omega, t}(\cdot | s, p, a)} \log \frac{q_\nu^+(y \mid s, p, a, O)}{q(y)}. \end{aligned} \quad (24)$$

Proof. For fixed s, p, a , write the joint mixture of (Y, O) under ν as $w_{y,o}$ and the observation mixture as $w_o = \sum_y w_{y,o}$. The information and reference term can be written as

$$\mathcal{I}_\nu(Y; O \mid s, p, a) + D_{\text{KL}}(q_\nu \| q) = \sum_{y,o} w_{y,o} \log \frac{w_{y,o}}{w_o q(y)}.$$

Differentiating with respect to the mass at (ω, y) gives

$$\mathbb{E}_{O \sim P_{\omega, t}(\cdot | s, p, a)} \log \frac{q_\nu^+(y \mid s, p, a, O)}{q(y)},$$

because the +1 terms from the numerator and denominator cancel. Averaging over $a \sim p$ and adding the derivative of the linear loss term proves (23). Pairing the gradient with $\delta_{(\omega, y)} - \nu$ cancels the posterior-averaged terms and gives (24). $\square$

4.3 Exact AIR identity and model-index specialization

We now present simplified statements and alternative proofs of several results that are central to the AIR/MAIR literature (Xu and Zeevi, 2025; Liu et al., 2025, 2026). These results are regret identities, not merely upper bounds, but the identity has a specific hypothesis: the next index marginal must be the posterior index marginal generated by the pair belief ν_t that appears in the AIR functional. Under this ν -generated update, Lemma 4.3 gives the one-step AIR bracket and the coordinate logarithm telescopes. Other reference updates, such as confidence-to-belief or exponential-weights updates, may yield useful log-potential certificates, but they are not the AIR identity unless they can be represented by an admissible pair belief and the corresponding posterior-index update. Thus AIR is the exact posterior-scoring algebra inside the Bellman program; more general log-potential bounds are certified by separate calibration or relaxation arguments.

Theorem 4.4 (Exact indexed AIR regret identity). *Let an algorithm generate, at each reached state S_t , a current reference index marginal $q_t \in \text{int } \Delta(\mathcal{Y})$, a pair belief $\nu_t \in \Delta(\Omega \times \mathcal{Y})$, and a decision law $p_t \in \Delta(\mathcal{A}_t(S_t))$. After sampling $A_t \sim p_t$ and observing O_t , update the reference marginal by*

$$q_{t+1}(\cdot) = q_{\nu_t}^+(\cdot \mid S_t, p_t, A_t, O_t). \quad (25)$$

Fix a pair (ω^, y^*) with $q_t(y^*) > 0$ along the analyzed trajectory. Then, for every $\gamma > 0$,*

$$\mathbb{E}_{\omega^*}^{\text{Alg}} \sum_{t=1}^T \ell_{\omega^*, y^*}(S_t, p_t) = \gamma \mathbb{E}_{\omega^*}^{\text{Alg}} \log \frac{q_{T+1}(y^*)}{q_1(y^*)} + \mathbb{E}_{\omega^*}^{\text{Alg}} \sum_{t=1}^T B_{q_t, \gamma}^{\text{AIR}}(S_t, p_t, \nu_t; \omega^*, y^*). \quad (26)$$

For the original fixed-truth regret, take $y^ = \chi(\omega^*)$ and $\ell_{\omega^*, y^*} = \ell_{\omega^*}$. The expectation is over action randomization, observations, and any exogenous algorithmic randomness.*

Proof. Condition on H_{t-1} . By the update rule (25),

$$\mathbb{E} \left[\log \frac{q_{t+1}(y^*)}{q_t(y^*)} \middle| H_{t-1} \right] = \mathbb{E}_{a \sim p_t, O \sim P_{\omega^*, t}(\cdot \mid S_t, p_t, a)} \log \frac{q_{\nu_t}^+(y^* \mid S_t, p_t, a, O)}{q_t(y^*)}.$$

Combining this equality with (24) gives the one-step identity between expected loss, bracket, and log-ratio increment. Summing over t telescopes the logarithm. $\square$

Danskin, posterior scoring, and why the AIR bracket is not accidental The short algebraic proofs above are the fixed-truth version of the variational argument behind AIR. In the Bellman-sufficient notation, fix a state s , an action law p , and the current state marginal $q_s \in \Delta(\mathcal{Y})$ for the index. Let

$$\nu \in \Delta(\Omega \times \mathcal{Y})$$

be an admissible pair belief, usually with $Y = \chi(\Omega)$ and Y -marginal q_s . Given a prediction rule

$$Q : (s, p, a, o) \mapsto \Delta(\mathcal{Y}),$$

consider the posterior-scoring objective

$$\mathcal{J}_\gamma(s, p, \nu, Q; q_s) := \mathbb{E}_{\substack{(\Omega, Y) \sim \nu \\ A \sim p \\ O \sim P_{\Omega, s, p, A}}} \left[\ell_{\Omega, Y}(s, p, A) - \gamma \log \frac{Q(s, p, A, O)(Y)}{q_s(Y)} \right].$$

Here q_s is held fixed as part of the current state. If the analysis also optimizes over the index marginal itself, then that marginal is part of the Bellman control or enlarged state; the local Danskin statement is applied after fixing its current value.

For fixed (s, p, ν, q_s) , the logarithmic score is convex in Q and is minimized, pointwise in (s, p, a, o) , by the posterior index marginal generated by the same pair belief ν :

$$Q_\nu(s, p, a, o) = q_\nu^+(\cdot \mid s, p, a, o) := \nu(Y \in \cdot \mid s, p, a, O = o),$$

with the usual regular-conditional interpretation, or equivalently the Bayes formula under a dominated observation model. The optimized value

$$\mathcal{A}_\gamma(s, p, \nu; q_s) := \inf_Q \mathcal{J}_\gamma(s, p, \nu, Q; q_s)$$

is exactly the indexed AIR one-step objective: it is immediate indexed loss minus γ times the posterior log gain of the index.

For fixed Q and fixed q_s , the unoptimized objective is affine in the pair belief ν . Hence the optimized value is concave in ν , being the infimum of affine functions. Under the standard compactness, support, and differentiability conditions needed to apply Danskin's theorem, the directional derivative of $\mathcal{A}_\gamma$ is obtained by freezing the optimizer $Q_\nu = q_\nu^+$. Therefore the pointwise gradient in the direction of a fixed truth (ω^*, y^*) is precisely

$$\ell_{\omega^*, y^*}(s, p) - \gamma \mathbb{E}_{O \sim P_{\omega^*, s, p, A}}^{A \sim p} \log \frac{q_\nu^+(y^* | s, p, A, O)}{q_s(y^*)},$$

which is the AIR gradient bracket used above. Thus the bracket is not an accidental cancellation: posterior scoring selects the Bayes index predictor, Danskin differentiates through that selected predictor, and the coordinate log score telescopes along the maintained reference update.

The notation in [Xu and Zeevi \(2025, Section 5\)](#) is the action-index specialization $Y = A^*(\Omega)$. The environment-index specialization $Y = \Omega$ gives the MAIR or model-index form. In the stationary-posterior case, ν is the current posterior over (Ω, Y) and $q_s = \nu_Y$; in robust AIR, AMS, or EBO, ν may be a selected or optimized admissible belief, and validity comes from the corresponding Bellman bracket or calibration certificate. The same concavity-convexity structure also explains the Nash-equilibrium statement in [Xu and Zeevi \(2025, Lemma 5.2\)](#), although the formal proof still has to check compactness, support, and boundary conditions.

Identity versus lower bound. The identity is tightest when the final coordinate potential is retained. Rearranging (26) gives

$$\mathbb{E}_{\omega^*}^{\text{Alg}} \sum_{t=1}^T \ell_{\omega^*, y^*}(S_t, p_t) = \gamma \log \frac{1}{q_1(y^*)} - \gamma \mathbb{E}_{\omega^*}^{\text{Alg}} \log \frac{1}{q_{T+1}(y^*)} + \mathbb{E}_{\omega^*}^{\text{Alg}} \sum_{t=1}^T B_{q_t, \gamma}^{\text{AIR}}(S_t, p_t, \nu_t; \omega^*, y^*).$$

Dropping the nonpositive terminal term $-\gamma \mathbb{E} \log(1/q_{T+1}(y^*))$ gives the usual upper value and loses exactly the remaining coordinate code length. This observation does not itself give a minimax lower bound; the lower side still requires a Bellman-Fano or ghost-quantile value. It explains why the same logarithmic coordinate is the right quantity on both sides: the upper proof pays the initial code length only after discarding the final unresolved code length, while the lower proof certifies how much indexed entropy must remain under ghost histories.

Stationary-posterior specialization. In the exact posterior-indexed lift of Definition 2.3, the pair belief is the current posterior law of $(\Omega, \chi(\Omega))$ and q_t is its index marginal. Then $q_{\nu_t}^+$ is the exact posterior index update and the one-step fixed-truth log gain can also be written in predictive-law form. For $y = \chi(\omega)$,

$$J_\chi(s, p; \omega) := \mathbb{E}_{a \sim p, O \sim P_{\omega, t}(\cdot | s, p, a)} \log \frac{q_{\nu_s}^+(y | s, p, a, O)}{q_s(y)}. \quad (27)$$

Equivalently, when the relevant laws are dominated,

$$J_\chi(s, p; \omega) = \mathbb{E}_{a \sim p} \left[D_{\text{KL}}(P_{\omega, s, p, a} \| P_{s, p, a}) - D_{\text{KL}}(P_{\omega, s, p, a} \| P_{s, p, a}^{\chi(\omega)}) \right]. \quad (28)$$

Posterior averaging of J_χ over ω conditional on $Y = y$, and then over $y \sim q_s$, recovers $\mathcal{I}_\chi(s, p)$. Thus J_χ is a fixed-truth coordinate log gain, whereas $\mathcal{I}_\chi$ is its posterior-averaged information counterpart.

For $\gamma > 0$, the exact-posterior fixed-truth indexed bracket is

$$B_{\chi,\gamma}(s, p; \omega) := \ell_\omega(s, p) - \gamma J_\chi(s, p; \omega). \quad (29)$$

It is the specialization of (24) obtained by taking the pair belief to be the current posterior over $(\Omega, \chi(\Omega))$ and the reference marginal to be q_s .

Environment-index specialization. If $Y = \Omega$, the pair belief is concentrated on the graph $y = \omega$. Write the current environment belief as μ and the reference model marginal as ρ . Then (21) becomes

$$\text{MAIR}_{\rho,\gamma}(s, p, \mu) := \mathbb{E}_{\omega \sim \mu} \ell_\omega(s, p) - \gamma \mathbb{E}_{a \sim p} I_\mu(\Omega; O \mid s, p, a) - \gamma D_{\text{KL}}(\mu \parallel \rho). \quad (30)$$

The fixed-truth bracket is

$$B_{\rho,\gamma}^{\text{MAIR}}(s, p, \mu; \omega^*) = \ell_{\omega^*}(s, p) - \gamma \mathbb{E}_{a \sim p} D_{\text{KL}}(P_{\omega^*,s,p,a} \parallel P_{\mu,s,p,a}) - \gamma \log \frac{\mu(\omega^*)}{\rho(\omega^*)}. \quad (31)$$

If ρ_{t+1} is the Bayes posterior update of μ_t after (S_t, p_t, A_t, O_t) , then

$$\mathbb{E}_{\omega^*}^{\text{Alg}} L_{\omega^*}(H_T) = \gamma \mathbb{E}_{\omega^*}^{\text{Alg}} \log \frac{\rho_{T+1}(\omega^*)}{\rho_1(\omega^*)} + \mathbb{E}_{\omega^*}^{\text{Alg}} \sum_{t=1}^T B_{\rho_t,\gamma}^{\text{MAIR}}(S_t, p_t, \mu_t; \omega^*). \quad (32)$$

The stationary or current-posterior choice $\mu_t = \rho_t$ removes the explicit density-ratio correction in (31) and gives the KL-DEC offset. In this specialization the potential is the model-coordinate log posterior. In action-index AIR it is instead the log posterior of the optimal action or policy. The latter is the sharper choice whenever the decision problem only requires the index and not the whole environment.

Lemma 4.5 (MAIR gradient bracket and three canonical specializations). *Assume the observation laws are dominated and the displayed derivatives exist. For fixed s, p, ρ, μ , up to an additive constant on the probability simplex,*

$$\frac{\partial}{\partial \mu(\omega)} \text{MAIR}_{\rho,\gamma}(s, p, \mu) = \ell_\omega(s, p) - \gamma \log \frac{\mu(\omega)}{\rho(\omega)} - \gamma \mathbb{E}_{a \sim p} D_{\text{KL}}(P_{\omega,s,p,a} \parallel P_{\mu,s,p,a}).$$

Consequently, for every ω^* ,

$$\begin{aligned} & \text{MAIR}_{\rho,\gamma}(s, p, \mu) + \langle \nabla_\mu \text{MAIR}_{\rho,\gamma}(s, p, \mu), \delta_{\omega^*} - \mu \rangle \\ &= \ell_{\omega^*}(s, p) - \gamma \mathbb{E}_{a \sim p} D_{\text{KL}}(P_{\omega^*,s,p,a} \parallel P_{\mu,s,p,a}) - \gamma \log \frac{\mu(\omega^*)}{\rho(\omega^*)}. \end{aligned} \quad (33)$$

This bracket has three common specializations.

1. If $\mu = \rho$, the logarithmic correction vanishes and the bracket becomes the KL-DEC offset

$$\ell_{\omega^*}(s, p) - \gamma \mathbb{E}_{a \sim p} D_{\text{KL}}(P_{\omega^*,s,p,a} \parallel P_{\mu,s,p,a}). \quad (34)$$

2. If $\bar{\mu} \in \arg \max_{\mu \in \Delta(\Omega)} \text{MAIR}_{\rho,\gamma}(s, p, \mu)$ is an interior maximizer, then

$$\text{MAIR}_{\rho,\gamma}(s, p, \bar{\mu}) + \langle \nabla_\mu \text{MAIR}_{\rho,\gamma}(s, p, \bar{\mu}), \delta_{\omega^*} - \bar{\mu} \rangle \leq \text{MAIR}_{\rho,\gamma}(s, p, \bar{\mu}). \quad (35)$$

This is the MAMS/MEBO bracket control.

3. Let $p_{\omega,a} = dP_{\omega,s,p,a}/d\nu_{s,p,a}$ and define, for an auxiliary pair $(\bar{a}, \bar{o})$,

$$\frac{d\mu_{\bar{a},\bar{o}}^{1/2}}{d\rho}(\omega) := \frac{\sqrt{p_{\omega,\bar{a}}(\bar{o})}}{\int \sqrt{p_{\omega',\bar{a}}(\bar{o})} \rho(d\omega')}. \quad (36)$$

Let $h^2(P, Q) := 1 - \int \sqrt{dP dQ}$. If $\bar{a} \sim p$ and $\bar{o} \sim P_{\omega^*,s,p,\bar{a}}$, then

$$\begin{aligned} & \mathbb{E}_{\bar{a},\bar{o}} \left[\text{MAIR}_{\rho,\gamma}(s, p, \mu_{\bar{a},\bar{o}}^{1/2}) + \langle \nabla_{\mu} \text{MAIR}_{\rho,\gamma}(s, p, \mu_{\bar{a},\bar{o}}^{1/2}), \delta_{\omega^*} - \mu_{\bar{a},\bar{o}}^{1/2} \rangle \right] \\ & \leq \ell_{\omega^*}(s, p) - \gamma \mathbb{E}_{\bar{a} \sim p} \mathbb{E}_{\omega \sim \rho} h^2(P_{\omega^*,s,p,\bar{a}}, P_{\omega,s,p,a}). \end{aligned} \quad (37)$$

Thus the square-root posterior turns the MAIR bracket into a Hellinger DEC offset.

Proof. The derivative and the first two specializations are exactly the environment-index specialization of Lemma 4.3. For the square-root claim, fix $\bar{a}$ and write

$$Z_{\bar{a}}(\bar{o}) := \int \sqrt{p_{\omega',\bar{a}}(\bar{o})} \rho(d\omega').$$

Then

$$\log \frac{d\mu_{\bar{a},\bar{o}}^{1/2}}{d\rho}(\omega^*) = \frac{1}{2} \log p_{\omega^*,\bar{a}}(\bar{o}) - \log Z_{\bar{a}}(\bar{o}).$$

Under $\bar{o} \sim P_{\omega^*,s,p,\bar{a}}$, Jensen's inequality gives

$$\begin{aligned} \mathbb{E} \log \frac{d\mu_{\bar{a},\bar{o}}^{1/2}}{d\rho}(\omega^*) &= -\mathbb{E} \log \frac{Z_{\bar{a}}(\bar{o})}{\sqrt{p_{\omega^*,\bar{a}}(\bar{o})}} \\ &\geq -\log \int \sqrt{p_{\omega^*,\bar{a}}(o)} Z_{\bar{a}}(o) d\nu_{s,p,\bar{a}}(o) \\ &= -\log (1 - \mathbb{E}_{\omega \sim \rho} h^2(P_{\omega^*,s,p,\bar{a}}, P_{\omega,s,p,\bar{a}})) \\ &\geq \mathbb{E}_{\omega \sim \rho} h^2(P_{\omega^*,s,p,\bar{a}}, P_{\omega,s,p,\bar{a}}). \end{aligned}$$

Averaging over $\bar{a} \sim p$, substituting into (33), and dropping the nonnegative KL term proves (37). $\square$

5 Upper bounds: one identity and four algorithm families

We first isolate the fixed-truth log-potential telescope and then specify when it is an AIR/MAIR identity. The potential is the logarithmic score of the maintained index marginal. For AIR/MAIR identities, the next marginal is not arbitrary: it is the posterior index marginal produced by a specified pair belief ν over (Ω, Y) . Exact Bayes and stationary-posterior MAIR are special cases of this construction. Confidence-to-belief and exponential-weights procedures can also be used as logarithmic reference updates, but then the displayed Bellman bracket is a certificate for that update rather than the AIR gradient identity unless an admissible pair-belief representation has been verified.

Fix a truth ω^* and write $y^* = \chi(\omega^*)$. At state s , the log potential uses the current reference marginal q_s on the index. A local algorithmic control u specifies an action law $p_u \in \Delta(\mathcal{A}_t(s))$ and a state-measurable reference-index update

$$q_u^+(\cdot \mid s, a, o) \in \Delta(\mathcal{Y}).$$

In the AIR/MAIR specialization, this update is generated by a pair belief $\nu_u \in \Delta(\Omega \times \mathcal{Y})$, namely

$$q_u^+(\cdot \mid s, a, o) = q_{\nu_u}^+(\cdot \mid s, p_u, a, o),$$

with q_ν^+ defined in (20). Exact Bayes and stationary-posterior MAIR are ν -generated updates. Confidence-to-belief and exponential-weights procedures may also define logarithmic reference updates, but then the theorem uses the corresponding state-measurable update directly; such a certificate should not be called the AIR gradient identity unless an admissible pair-belief representation has been verified.

The fixed-truth reference log gain used by the Bellman program is

$$\mathbf{G}_\chi(s, u; \omega^\star) := \mathbb{E}_{a \sim p_u, O \sim P_{\omega^\star, t}(\cdot \mid s, p_u, a)} \log \frac{q_u^+(y^\star \mid s, a, O)}{q_s(y^\star)}. \quad (38)$$

When $u = (p, \nu)$ is represented by an AIR pair belief, we also write $\mathbf{G}_\chi(s, p, \nu; \omega^\star)$. In that case $q_u^+ = q_\nu^+$, and $\mathbf{G}_\chi$ is the fixed-truth AIR posterior-coordinate log gain. When ν is the exact posterior pair law, this is $J_\chi(s, p; \omega^\star)$ from (27). Thus the log potential is always the coordinate log score of a specified reference index marginal, but the AIR/MAIR regret identity additionally requires that the next marginal be the ν -posterior coordinate associated with the same pair belief used in the AIR functional.

For any predictable continuation potential V_t on states and any coefficient $\gamma > 0$, define the specialized fixed-truth logarithmic potential

$$\Phi_t^{V, \gamma}(s; y^\star) := V_t(s) + \gamma \log \frac{1}{q_s(y^\star)}. \quad (39)$$

The corresponding log-potential Bellman bracket is

$$\mathfrak{B}_\chi^{V, \gamma}(s, u; \omega^\star) := \ell_{\omega^\star}(s, p_u) + \mathbb{E}_{\omega^\star}[V_{t+1}(S^+) \mid s, u] - V_t(s) - \gamma \mathbf{G}_\chi(s, u; \omega^\star). \quad (40)$$

Here $A \sim p_u$, $O \sim P_{\omega^\star, t}(\cdot \mid s, p_u, A)$, the reference part of the next state is updated by $q_u^+(\cdot \mid s, A, O)$, and any other state variables are updated by the rule specified by u . In the AIR specialization $u = (p, \nu)$, this bracket is written $\mathfrak{B}_\chi^{V, \gamma}(s, p, \nu; \omega^\star)$. The identity behind the upper section is the telescope

$$\begin{aligned} \mathbb{E}_{\omega^\star} L_{\omega^\star}(H_T) &= V_1(s_1) - \mathbb{E}_{\omega^\star} V_{T+1}(S_{T+1}) + \gamma \log \frac{1}{q_1(y^\star)} - \gamma \mathbb{E}_{\omega^\star} \log \frac{1}{q_{T+1}(y^\star)} \\ &\quad + \mathbb{E}_{\omega^\star} \sum_{t=1}^T \mathfrak{B}_\chi^{V, \gamma}(S_t, u_t; \omega^\star). \end{aligned} \quad (41)$$

Retaining the terminal term gives the sharp identity for the analyzed algorithm. The common upper bound drops the nonpositive term $-\gamma \mathbb{E} \log(1/q_{T+1}(y^\star))$; the overestimate from this simplification is exactly the remaining coordinate code length. This is an upper identity, not a minimax lower bound. A Bellman–Fano lower bound is still a separate argument, but it is naturally matched to the same coordinate because it certifies how much index entropy can remain under low-regret ghost histories.

Thus an upper bound is a Bellman supersolution for this log-potential bracket. The exact indexed AIR Bellman program optimizes dynamically over the state, action law, and any algorithmically chosen reference-update rule. If that update rule is represented by a pair belief, the pair belief is an algorithmic prediction variable. By contrast, AMS/EBO places candidate beliefs under a robust maximization and uses a saddle or first-order condition to certify the same fixed-truth AIR bracket. UCB certifies the bracket by calibration and optimism, while E2D drops or bounds the continuation to obtain a one-step offset.

Theorem 5.1 (Generic fixed-truth indexed AIR/MAIR upper bound). *Fix ω^* and $y^* = \chi(\omega^*)$. Assume $q_1(y^*) > 0$ and that the reference update remains positive on y^* along the analyzed trajectory. If an algorithm chooses a decision law p_t and a state-measurable prediction belief ν_t so that the reference update is $q_{t+1} = q_{\nu_t}^+(\cdot \mid S_t, p_t, A_t, O_t)$ and, for every reached state,*

$$\mathfrak{B}_\chi^{V, \gamma}(S_t, p_t, \nu_t; \omega^*) \leq \varepsilon_t,$$

then

$$\mathbb{E}_{\omega^*}^{\text{Alg}} L_{\omega^*}(H_T) \leq V_1(s_1) + \gamma \log \frac{1}{q_1(y^*)} - \mathbb{E}_{\omega^*}^{\text{Alg}} V_{T+1}(S_{T+1}) - \gamma \mathbb{E}_{\omega^*}^{\text{Alg}} \log \frac{1}{q_{T+1}(y^*)} + \mathbb{E}_{\omega^*}^{\text{Alg}} \sum_{t=1}^T \varepsilon_t. \quad (42)$$

Proof. Taking the conditional expectation of $\gamma \log(1/q_{t+1}(y^*)) - \gamma \log(1/q_t(y^*))$ under $A \sim p$ and $O \sim P_{\omega^*, t}(\cdot \mid s, p, A)$, with $q_{t+1} = q_{\nu_t}^+(\cdot \mid s, p, A, O)$, gives $-\gamma \mathbf{G}_\chi(s, p, \nu; \omega^*)$. Substituting the bracket inequality into the telescoping identity (41) proves (42). $\square$

It is useful to keep one concrete state specialization in mind. A fixed estimation procedure maps histories to a reference update. In AIR form the update is represented by a pair belief ν_t over (Ω, Y) ; in an exact model-posterior specialization it is a belief over environments with $Y = \chi(\Omega)$. The retained state includes the current reference index marginal

$$q_t \in \Delta(\mathcal{Y}), \quad (43)$$

and, when the update is posterior based, the next marginal is $q_{t+1} = q_{\nu_t}^+(\cdot \mid S_t, p_t, A_t, O_t)$. In a well-specified Bayes analysis, ν_t is the exact posterior over $(\Omega, \chi(\Omega))$. In a frequentist analysis, the state may instead contain a calibrated algorithmic belief, confidence object, or exponential-weights density. Such an object is valid in the AIR identity only when it induces an admissible pair belief and posterior-coordinate update; otherwise it is a separate log-score certificate whose validity must be proved through calibration or domination. Posterior averaging of the coordinate potential gives

$$\mathbb{E}_{Y \sim \nu_\chi} \gamma \log \frac{1}{q_s(Y)} = \gamma H(\nu_\chi) + \gamma D_{\text{KL}}(\nu_\chi \| q_s), \quad (44)$$

and for a finite index set the same logarithmic scoring rule has the KL-dual log-partition form

$$\Psi_\gamma(q, z) := \gamma \log \sum_{y \in \mathcal{Y}} q(y) e^{zy/\gamma} = \sup_{r \in \Delta(\mathcal{Y})} \{ \langle r, z \rangle - \gamma D_{\text{KL}}(r \| q) \}. \quad (45)$$

The coordinate log loss, its posterior average, and the dual log partition are three representations of the same KL/logarithmic scoring rule. They are not interchangeable in a proof: fixed-truth guarantees use (39), posterior-averaged Bayes identities use (44), and AMS/EBO relaxations use (45).

In the model-index specialization, write the posterior/reference belief as μ_t . If the maintained reference update is exact Bayes, then $\mathbf{G}_\chi = J_\chi$ and the fixed-truth MAIR information increment is

$$J_t^{\text{MAIR}}(p; \omega^*) := \mathbb{E}_{a \sim p} D_{\text{KL}}(P_{\omega^*, t}(\cdot \mid S_t, a) \| P_{\mu_t, t}(\cdot \mid S_t, a)). \quad (46)$$

The action-index AIR version is identical with the index $Y = A^*$, the reference marginal q_t , and the AIR posterior coordinate $q^+(a^* \mid a, o)$ in place of the model posterior.

5.1 Information-potential Bellman programming

The exact frequentist upper algorithm is an indexed AIR log-penalized Bellman program on calibrated state/index representations. Let $\mathfrak{C}_t(s)$ be the set of environments that the algorithmic state regards as admissible at state s . Let $\mathfrak{U}_t(s)$ be the local set of admissible algorithmic controls. A control $u \in \mathfrak{U}_t(s)$ specifies an action law $p_u \in \Delta(\mathcal{A}_t(s))$ and a reference-update rule. When this update is represented by a pair belief, write that belief as ν_u and set $q^+ = q_{\nu_u}^+$. If the pair belief is fixed by the state, then u is simply the action law together with this fixed update. The robust AMS/EBO belief variable is not this algorithmic control; it is introduced later under a supremum to certify the bracket. For $\gamma > 0$, define

$$W_{T+1}^\gamma(s) = 0, \quad W_t^\gamma(s) := \inf_{u \in \mathfrak{U}_t(s)} \sup_{\omega \in \mathfrak{C}_t(s)} \left\{ \ell_\omega(s, p_u) + \mathbb{E}_\omega[W_{t+1}^\gamma(S^+) \mid s, u] - \gamma G_\chi(s, u; \omega) \right\}. \quad (47)$$

A measurable selector attaining the infimum in (47) is the information-penalized Bellman dynamic-programming algorithm. In the common AIR specialization, $u = (p, \nu)$ and $G_\chi(s, u; \omega) = G_\chi(s, p, \nu; \omega)$; in exact Bayes, ν is fixed by the posterior state; in a non-AIR log-score certificate, u specifies the state-measurable reference update directly. This construction builds on the principle of admissible relaxations (Rakhlin et al., 2012), which organize learning through dynamic programming over loss states, and is particularly close to the partial-information relaxation viewpoint of Rakhlin and Sridharan (2016), where estimator-like state variables and potential terms quantify uncertainty. It can also be viewed as a sequential analogue of classical minimum-complexity, information-risk, and Gibbs-type estimation in fixed experiments (Barron and Cover, 1991; Yang and Barron, 1999; Zhang, 2006). The key new feature is that a generic logarithmic mass penalty is propagated through a controlled Bellman recursion, rather than applied only once to a terminal estimator, and the recursion is carried by a Bellman-sufficient state together with an explicit information index.

Theorem 5.2 (Frequentist log-penalized Bellman upper bound). *Fix $\Omega_0 \subseteq \Omega$ and $\gamma > 0$. Suppose the reference update and the comparison sets $\mathfrak{C}_t$ are calibrated for Ω_0 in the following sense: for every $\omega^* \in \Omega_0$, on the calibration event $\mathcal{E}$, $\omega^* \in \mathfrak{C}_t(S_t)$ whenever round t is reached. Suppose the algorithm chooses a local control $u_t \in \mathfrak{U}_t(S_t)$, writes $p_t = p_{u_t}$ for its action law, updates the index marginal by the reference-update rule contained in u_t , and satisfies*

$$\sup_{\omega \in \mathfrak{C}_t(S_t)} \left\{ \ell_\omega(S_t, p_{u_t}) + \mathbb{E}_\omega[W_{t+1}^\gamma(S^+) \mid S_t, u_t] - \gamma G_\chi(S_t, u_t; \omega) \right\} \leq W_t^\gamma(S_t) + \varepsilon_t.$$

Then, for every fixed truth $\omega^ \in \Omega_0$ with $y^* = \chi(\omega^*)$ and $q_1(y^*) > 0$, provided the reference update remains positive on y^* on the calibrated trajectory,*

$$\mathbb{E}_{\omega^*}^{\text{Alg}} L_{\omega^*}(H_T) \leq W_1^\gamma(s_1) + \gamma \log \frac{1}{q_1(y^*)} + \mathbb{E}_{\omega^*}^{\text{Alg}} \sum_{t=1}^T \varepsilon_t + \text{cal}_T(\omega^*), \quad (48)$$

provided the index is finite or countable, where

$$\text{cal}_T(\omega^*) := \mathbb{E}_{\omega^*}[L_{\omega^*}(H_T)^+ \mathbb{1}\{\mathcal{E}^c\}].$$

If calibration holds surely, then $\text{cal}_T(\omega^) = 0$; if the positive part of the cumulative loss is bounded by B_T , then $\text{cal}_T(\omega^*) \leq B_T \mathbb{P}_{\omega^*}(\mathcal{E}^c)$. The common per-round bound $0 \leq \ell_t \leq L$ is only the special case $B_T = LT$.*

Proof. On the calibration event, the fixed truth is one of the environments in the robust supremum. Therefore the displayed approximate minimization implies

$$\ell_{\omega^*}(S_t, p_{u_t}) + \mathbb{E}_{\omega^*}[W_{t+1}^\gamma(S^+) \mid S_t, u_t] - W_t^\gamma(S_t) - \gamma G_\chi(S_t, u_t; \omega^*) \leq \varepsilon_t.$$

Apply the log-potential telescope (41) with $V_t = W_t^\gamma$. Since $W_{T+1}^\gamma = 0$ and the terminal index log loss is nonnegative for a finite or countable index, the result follows on $\mathcal{E}$. Splitting the expectation according to $\mathcal{E}$ and $\mathcal{E}^c$ and upper-bounding the failure contribution by the positive part gives the calibration term. $\square$

The posterior-averaged Bayesian recursion is a specialization rather than the primitive fixed-truth statement. If the robust comparison set is replaced by the current posterior law and one averages the coordinate identity over $Y \sim q_s$, then

$$\mathbb{E}[H_\chi(S_{t+1}) \mid s, p] - H_\chi(s) = -\mathcal{I}_\chi(s, p).$$

For a fixed multiplier $\gamma > 0$, define the Bayes information-regularized value

$$U_{T+1}^\gamma \equiv 0, \quad U_t^\gamma(s) := \inf_{p \in \Delta(\mathcal{A}_t(s))} \{ \ell(s, p) + \mathbb{E}[U_{t+1}^\gamma(S_{t+1}) \mid s, p] - \gamma \mathcal{I}_\chi(s, p) \}. \quad (49)$$

Equivalently,

$$\Phi_t^\gamma(s) := U_t^\gamma(s) + \gamma H_\chi(s) \quad (50)$$

solves the admissible-relaxation recursion

$$\Phi_t^\gamma(s) = \inf_{p \in \Delta(\mathcal{A}_t(s))} \{ \ell(s, p) + \mathbb{E}[\Phi_{t+1}^\gamma(S_{t+1}) \mid s, p] \}, \quad \Phi_{T+1}^\gamma(s) = \gamma H_\chi(s). \quad (51)$$

Theorem 5.3 (Posterior-averaged information-potential upper bound). *Assume the indexed Bellman representation is exact under the Bayesian mixture law and $H_\mu(Y) < \infty$. If a policy chooses p_t satisfying*

$$\ell(S_t, p_t) + \mathbb{E}[U_{t+1}^\gamma(S_{t+1}) \mid S_t, p_t] - \gamma \mathcal{I}_\chi(S_t, p_t) \leq U_t^\gamma(S_t) + \varepsilon_t,$$

then

$$\mathbb{E}L_\Omega(H_T) \leq U_1^\gamma(s_1) + \gamma I_\mu(Y; H_T) + \sum_{t=1}^T \varepsilon_t \leq U_1^\gamma(s_1) + \gamma H_\mu(Y) + \sum_{t=1}^T \varepsilon_t. \quad (52)$$

This is the posterior average of the fixed-truth coordinate telescope, not a pointwise replacement for Theorem 5.2.

Proof. Rearrange the displayed inequality and telescope U_t^γ . Proposition 4.2 identifies $\mathbb{E} \sum_t \mathcal{I}_\chi(S_t, p_t)$ with $I_\mu(Y; H_T)$, and $I_\mu(Y; H_T) \leq H_\mu(Y)$ gives the final bound. The equivalence with (51) follows from the posterior entropy-drop identity. $\square$

The minimax information-risk sandwich and the entropy/regret comparison are stated in Section 3. The rest of this upper-bound section develops the upper side of that sandwich: the fixed-truth coordinate telescope, the exact log-penalized Bellman program, and the main tractable relaxations.

5.2 UCB families: calibration plus optimism

An upper-confidence-bound (UCB) algorithm maintains a prediction and an uncertainty scale, builds a confidence band from them, and chooses an action that is optimistic within that band (Auer et al., 2002; Abbasi-Yadkori et al., 2011; Srinivas et al., 2010). We use the following notation throughout. The unmultiplied uncertainty scale is denoted by $\sigma_t(a)$ in the generic discussion and by $s_t(x)$ in the GP specialization below. The confidence half-width is

$$w_t(a) = \beta_t \sigma_t(a),$$

where β_t is the confidence multiplier. Calibration says that the fixed truth is contained in the band of radius $w_t(a)$; optimism says that the selected action maximizes the upper envelope. Together, in reward-maximization problems, these two facts give

$$r_{\omega^*}(s, a^*) \leq m_t(a^*) + w_t(a^*) \leq m_t(a_t) + w_t(a_t) \leq r_{\omega^*}(s, a_t) + 2w_t(a_t),$$

and hence the instantaneous regret is at most $2\beta_t \sigma_t(a_t)$. An elliptical-potential/log-determinant lemma then controls the realized sum of $\sigma_t^2(a_t)$. Thus UCB does not solve the AIR/MAIR bracket optimization directly; it certifies the relevant fixed-truth bracket after the optimistic action has been chosen.

Lemma 5.4 (Calibration and optimism imply bracket control). *Fix a truth ω^* and condition on an event $\mathcal{E}$ on which the algorithm's state is calibrated for that truth. Suppose that, for every reached round t , the action a_t chosen by the algorithm and a predictable uncertainty scale $\sigma_t(a) \geq 0$ satisfy*

$$\ell_{\omega^*}(S_t, a_t) \leq c_{\text{opt}} \beta_t \sigma_t(a_t), \quad (53)$$

where c_{opt} is a numerical optimism constant; in the usual two-sided reward-confidence proof above, $c_{\text{opt}} = 2$. Then for every $\gamma > 0$,

$$\ell_{\omega^*}(S_t, a_t) - \gamma \sigma_t^2(a_t) \leq \frac{c_{\text{opt}}^2 \beta_t^2}{4\gamma}. \quad (54)$$

Consequently, for the deterministic-action fixed-truth bracket in (29),

$$B_{\chi, \gamma}(S_t, \delta_{a_t}; \omega^*) \leq \frac{c_{\text{opt}}^2 \beta_t^2}{4\gamma} + \gamma (\sigma_t^2(a_t) - J_{\chi}(S_t, \delta_{a_t}; \omega^*)), \quad (55)$$

where the fixed-truth index log gain is defined in (27). If, in addition, the log-determinant/elliptical-potential argument gives a realized bound

$$\sum_{t=1}^T \sigma_t^2(a_t) \leq c_{\text{sn}} G_T,$$

for an information telescope G_T , then summing (55) reduces UCB regret control to the comparison between G_T and the fixed-truth log-gain telescope.

Proof. The optimism/calibration hypothesis (53) and Young's inequality give

$$c_{\text{opt}} \beta_t \sigma_t(a_t) \leq \frac{c_{\text{opt}}^2 \beta_t^2}{4\gamma} + \gamma \sigma_t^2(a_t).$$

This proves (54). Subtracting $\gamma J_{\chi}(S_t, \delta_{a_t}; \omega^*)$ from both sides and using the definition of $B_{\chi, \gamma}$ proves (55). The final sentence is the result of summing the displayed inequality over time. $\square$

The correction $\sigma_t^2(a_t) - J_\chi(S_t, \delta_{a_t}; \omega^*)$ is not a second estimation cost. It is the algebraic difference between the tractable self-normalized variance proxy and the fixed-truth posterior log gain appearing in the bracket. In linear-UCB and GP-UCB proofs, the log-determinant or elliptical-potential argument is precisely the certificate that the realized sum of $\sigma_t^2(a_t)$ is controlled by an information telescope. The confidence multiplier β_t is separate: it is the price of converting calibrated uncertainty into regret through optimism. Therefore

$$\text{self-normalized calibration} + \text{optimism} \implies \text{fixed-truth AIR/MAIR bracket control.}$$

In this organizing perspective, the round-heterogeneous summation of unmultiplied uncertainty scales, usually controlled by an elliptical-potential or log-determinant argument, plays the role of a realized estimation-complexity telescope. The confidence multiplier enters instead through the regret-to-uncertainty conversion, and hence through the coefficient in the fixed-truth bracket. This separates two mechanisms that are often blurred in informal UCB discussions: posterior or confidence widths quantify the “estimation complexity” of the underlying parameter, as explained by [Lattimore \(2023\)](#), while optimism and calibration determine how that geometry pays for regret. The resulting unification reveals a common posterior-telescoping structure while keeping distinct the estimation telescope and the coefficient mechanisms behind UCB, E2D, and AMS/EBO.

5.3 E2D: robust one-step offset optimization

The estimation-to-decision (E2D) algorithms ([Foster et al., 2021](#)) choose p by solving a one-round robust DEC offset at the current time-state pair. In model-index notation the schematic KL form is

$$\inf_{p \in \Delta(\mathcal{A}_t(s))} \sup_{\omega \in \mathfrak{C}_t(s)} \{ \ell_\omega(s, p) - \gamma \mathbb{E}_{a \sim p} D_{\text{KL}}(P_{\omega, s, p, a} \| P_{\rho, s, p, a}) \}, \quad (56)$$

where ρ is a reference belief or reference model and $\mathfrak{C}_t(s)$ is a localized comparison set that contains the fixed truth on the calibration event. The notation is intentionally aligned with the log-penalized Bellman program. As [Lemma 4.5](#) illustrates, the E2D objective can be recovered from the one-step MAIR bracket: it uses the current feasible action set and localized comparison set, but replaces the dynamic continuation value by a one-step separation penalty. The same one-round minimax principle applies to the original Hellinger formulation and to constrained variants ([Foster et al., 2023](#)).

What E2D keeps is the immediate regret and a one-step statistical-separation penalty. What it drops, freezes, or upper bounds is the Bellman continuation term $\mathbb{E}_\omega[V_{t+1}(S^+) \mid s, p] - V_t(s)$, together with the exact coordinate reference log gain G_χ in (40). The stationary-posterior recovery (34) and the stationary-square-root-posterior recovery (37) make this reduction from the MAIR bracket explicit. Thus E2D is not an optimizer of the full Bellman potential unless one proves that the omitted continuation is dominated by the displayed one-step penalty. In an exact Bayesian calculation, posterior averaging can turn the KL penalty into a one-step mutual-information term. In a fixed-truth or frequentist calculation, the corresponding statement is the reference posterior-ratio identity for G_χ , or its exact-Bayes specialization J_χ , plus whatever calibration or localization is needed to compare $P_{\rho, s, p, a}$ with the reference predictive law in that identity. The safe interpretation is therefore: E2D is a one-step relaxation, re-solved at each state, of the fixed-truth log-posterior Bellman bracket.

5.4 AMS/EBO: robust convex belief optimization

The Exploration by Optimization (EBO) algorithm ([Lattimore and György, 2021](#); [Foster et al., 2022](#)) and the Adaptive Minimax Sampling (AMS) algorithm ([Xu and Zeevi, 2025](#); [Liu et al., 2025](#),

2026) should be understood as tractable robust relaxations of the log-penalized Bellman program, not as the same minimization over the algorithmic update variable in (47). AMS may be viewed as a constructive implementation of the EBO principle, replacing abstract functional-estimator optimization with an executable sampling rule. Work with an action or policy index $Y = \chi(\Omega)$. At state s , let $q \in \text{int } \Delta(\mathcal{Y})$ be the current reference marginal and let $\mathfrak{B}_t(s)$ be a calibrated convex set of candidate pair beliefs $\tilde{\nu}$ on (ω, y) , supported on $y = \chi(\omega)$. In this subsection the candidate belief is a robust comparison variable controlled by the inner maximization; after the maximizer $\bar{\nu}_t$ is selected, its posterior index update is used to certify the fixed-truth AIR bracket. The one-step robust AIR/EBO objective is the indexed AIR functional from (21), equivalently

$$\mathfrak{A}_{q,\gamma}(s, p, \nu) = \ell_\nu(s, p) - \gamma \mathcal{I}_\chi(s, p; \nu) - \gamma D_{\text{KL}}(\nu_\chi \| q), \quad (57)$$

where ν_χ is the index marginal, $\ell_\nu(s, p) = \mathbb{E}_{(\omega, y) \sim \nu} \ell_\omega(s, p)$, and $\mathcal{I}_\chi(s, p; \nu) = \mathbb{E}_{a \sim p} I_\nu(Y; O \mid s, p, a)$. The robust maximization over beliefs is essential:

$$p_t \in \arg \min_{p \in \Delta(\mathcal{A}_t(s))} \max_{\tilde{\nu} \in \mathfrak{B}_t(s)} \mathfrak{A}_{q_t, \gamma}(s, p, \tilde{\nu}), \quad \bar{\nu}_t \in \arg \max_{\tilde{\nu} \in \mathfrak{B}_t(s)} \mathfrak{A}_{q_t, \gamma}(s, p_t, \tilde{\nu}). \quad (58)$$

Without the inner $\max_{\tilde{\nu}}$, the display is only a posterior AIR calculation for a chosen belief, not a robust AMS/EBO rule.

The convex-analytic interpretation is the following. The fixed-truth proof uses the coordinate log loss $\gamma \log(1/q_t(y^*))$. Averaging this coordinate under a candidate belief gives $\gamma H(\nu_\chi) + \gamma D_{\text{KL}}(\nu_\chi \| q_t)$. When the next reference marginal is the posterior coordinate induced by ν , the expected change of this averaged potential is $-\gamma \mathcal{I}_\chi(s, p; \nu) - \gamma D_{\text{KL}}(\nu_\chi \| q_t)$. Equivalently, optimizing over possible next index laws yields the log-partition dual (45). Hence the EBO potential is the KL-dual log partition, while the fixed-truth AIR certificate is the coordinate posterior log loss; they coincide only through KL duality and the chosen update, not as identical scalar functions.

For a genuine convex belief optimization one must verify the convexity/concavity assumptions used by the saddle argument. In finite exponential-weights or admissible-relaxation settings this follows from the log-partition dual. In general model classes, mutual information need not be concave in an arbitrary belief when both the index marginal and the observation kernels vary. One must either verify concavity of $\tilde{\nu} \mapsto \mathfrak{A}_{q,\gamma}(s, p, \tilde{\nu})$ on $\mathfrak{B}_t(s)$, or replace it by a certified concave upper relaxation.

Lemma 5.5 (AMS/EBO saddle certifies the AIR fixed-truth bracket). *Assume the finite or dominated differentiability setting of Lemma 4.3. Fix (ω^*, y^*) with $y^* = \chi(\omega^*)$. Suppose $\tilde{\nu} \mapsto \mathfrak{A}_{q_t, \gamma}(s, p_t, \tilde{\nu})$ is concave on the convex set $\mathfrak{B}_t(s)$, $\bar{\nu}_t$ maximizes it as in (58), and the comparison direction $\delta_{(\omega^*, y^*)} - \bar{\nu}_t$ is admissible for $\mathfrak{B}_t(s)$; any failure of this containment is charged as a calibration error. If*

$$\max_{\tilde{\nu} \in \mathfrak{B}_t(s)} \mathfrak{A}_{q_t, \gamma}(s, p_t, \tilde{\nu}) \leq \varepsilon_t,$$

then the fixed-truth AIR bracket with coefficient γ satisfies

$$\ell_{\omega^*}(s, p_t) - \gamma \mathbb{E}_{a \sim p_t, O \sim P_{\omega^*, t}(\cdot | s, p_t, a)} \log \frac{\bar{\nu}_t(Y = y^* \mid s, p_t, a, O)}{q_t(y^*)} \leq \varepsilon_t. \quad (59)$$

If the reference update is $q_{t+1}(\cdot) = \bar{\nu}_t(Y \in \cdot \mid s, p_t, A_t, O_t)$ and the calibration assumptions hold along the trajectory, then Theorem 5.1 with $V_t \equiv 0$ gives

$$\mathbb{E}_{\omega^*} \text{Reg}_T \leq \gamma \log \frac{1}{q_1(y^*)} + \sum_{t=1}^T \varepsilon_t + \text{cal}_T \quad (60)$$

for finite or countable index sets. Averaging this coordinate bound under a Bayesian prior replaces $\log(1/q_1(Y))$ by the corresponding cross-entropy or by $H_\mu(Y)$ when $q_1 = \mu_\chi$.

Proof. The AIR gradient calculation in Lemma 4.3 identifies the fixed-truth bracket as the first-order value of $\mathfrak{A}_{q_t, \gamma}$ in the direction $\delta_{(\omega^*, y^*)} - \bar{\nu}_t$. Concavity and optimality of $\bar{\nu}_t$ make that first-order correction nonpositive for every admissible comparison direction. Hence the bracket is no larger than $\mathfrak{A}_{q_t, \gamma}(s, p_t, \bar{\nu}_t)$, which is at most the robust value and therefore at most ε_t . The regret bound is the fixed-truth log-potential telescope. $\square$

A dynamic AMS/EBO relaxation is obtained by adding a continuation term before the robust maximization:

$$\mathfrak{A}_{t, q, \gamma}^V(s, p, \tilde{\nu}) = \ell_{\tilde{\nu}}(s, p) + \mathbb{E}_{\tilde{\nu}} V_{t+1}(S^+) - V_t(s) - \gamma \mathcal{I}_\chi(s, p; \tilde{\nu}) - \gamma D_{\text{KL}}(\tilde{\nu}_\chi \| q).$$

This is a valid relaxation of the robust Bellman recursion (47) only after the same first-order or direct-bracket certificate is proved. The inner maximization over $\tilde{\nu}$ is a robust upper-bounding device; it does not contradict the Bellman-program minimization over algorithmic controls, because these variables play different roles. The standard one-step EBO display is the case $V_t \equiv 0$; the dynamic version is an admissible relaxation of the central log-penalized Bellman program, not an automatic consequence of the one-step objective.

Remark 5.6 (The one identity and the four families). *The AIR identity is the fixed-truth log-potential telescope (41) with a ν -generated posterior index update. The exact information-risk Bellman program is (47), where the minimization is over algorithmic controls and update rules. Posterior-averaged Bayes analysis is obtained by averaging the same coordinate identity, giving (49). UCB certifies the bracket through confidence and optimism. E2D replaces the continuation by a one-step separation penalty. AMS/EBO optimizes a KL-dual robust belief relaxation, with the required $\max_{\tilde{\nu}}$ over candidate beliefs appearing explicitly. Thus the common object is not a universal one-step coefficient but the logarithmic Bellman bracket on a sufficient state and a chosen index, with algorithmic prediction variables and robust comparison beliefs kept distinct.*

6 Lower bounds: reference histories and quantile indices

6.1 Ghost probability and true good probability

Fix an average-regret threshold $r \geq 0$. The posterior-reference ghost-good probability is

$$p_r^{\text{Alg}}(\mu, \chi) := \mathbb{P}_{Y \sim \mu_\chi, H'_T \sim \bar{\mathbb{P}}_\mu^{\text{Alg}}} (\bar{L}_\chi(Y, H'_T) \leq r), \quad (61)$$

where Y and H'_T are independent under the ghost law and μ_χ is the marginal law of Y . The true good probability is

$$q_r^{\text{Alg}}(\mu, \chi) := \mathbb{P}_{Y, H_T} (\bar{L}_\chi(Y, H_T) \leq r), \quad (62)$$

where (Y, H_T) are generated by $\Omega \sim \mu$ and $H_T \sim \mathbb{P}_\Omega^{\text{Alg}}$.

The ghost probability is algorithm-dependent. That dependence is intentional: it preserves the reference trajectory geometry instead of replacing it with a worst-case static small ball.

6.2 Reference-history quantile theorem

Let

$$\text{kl}(q, p) = q \log \frac{q}{p} + (1 - q) \log \frac{1 - q}{1 - p}$$

be binary KL divergence. For $p \in [0, 1]$ and $c \geq 0$, define

$$\psi(p, c) := \sup\{q \in [0, 1] : \text{kl}(q, p) \leq c\}.$$

Theorem 6.1 (Reference-history quantile indexed-information lower bound). *For every prior μ , index map χ , threshold $r \geq 0$, and algorithm Alg , assume that the posterior-reference process satisfies the exact posterior-indexed lift in the sense of Definition 2.3, and assume that the cumulative indexed loss satisfies $L_\chi(Y, H_T) \geq 0$ almost surely under the true mixture law,*

$$\text{kl}(q_r^{\text{Alg}}(\mu, \chi), p_r^{\text{Alg}}(\mu, \chi)) \leq C_\chi^{\text{Alg}}(\mu) = T\bar{C}_\chi^{\text{Alg}}(\mu). \quad (63)$$

Consequently,

$$\mathbb{E}_{\Omega \sim \mu} \mathbb{E}_{\mathbb{P}_\Omega^{\text{Alg}}} L_\Omega(H_T) \geq Tr \left[1 - \psi(p_r^{\text{Alg}}(\mu, \chi), T\bar{C}_\chi^{\text{Alg}}(\mu)) \right]. \quad (64)$$

In particular, if

$$T\bar{C}_\chi^{\text{Alg}}(\mu) \leq \text{kl}\left(\frac{1}{2}, p_r^{\text{Alg}}(\mu, \chi)\right), \quad (65)$$

then

$$\mathbb{E}_{\Omega \sim \mu} \mathbb{E}_{\mathbb{P}_\Omega^{\text{Alg}}} \bar{L}_\Omega(H_T) \geq \frac{r}{2}, \quad \mathbb{E}_{\Omega \sim \mu} \mathbb{E}_{\mathbb{P}_\Omega^{\text{Alg}}} L_\Omega(H_T) \geq \frac{Tr}{2}. \quad (66)$$

A convenient sufficient condition is

$$T\bar{C}_\chi^{\text{Alg}}(\mu) + \log 2 \leq \frac{1}{2} \log \frac{1}{p_r^{\text{Alg}}(\mu, \chi)}. \quad (67)$$

Proof. Consider the true joint law of (Y, H_T) and the ghost law of (Y, H'_T) . Both have the same marginal law of Y , while Y and H'_T are independent under the ghost law. Hence

$$D_{\text{KL}}(\mathcal{L}(Y, H_T) \| \mathcal{L}(Y, H'_T)) = I_\mu(Y; H_T).$$

Data processing applied to the indicator of the event $\{\bar{L}_\chi(Y, \cdot) \leq r\}$ gives

$$\text{kl}(q_r^{\text{Alg}}, p_r^{\text{Alg}}) \leq I_\mu(Y; H_T).$$

Proposition 4.2 identifies the right-hand side with $C_\chi^{\text{Alg}}(\mu) = T\bar{C}_\chi^{\text{Alg}}(\mu)$, proving (63). The definition of ψ gives $q_r^{\text{Alg}} \leq \psi(p_r^{\text{Alg}}, T\bar{C}_\chi^{\text{Alg}})$. By nonnegativity of the cumulative indexed loss and by (4),

$$\begin{aligned} \mathbb{E}_{\Omega, H_T} L_\Omega(H_T) &= \mathbb{E}_{Y, H_T} L_\chi(Y, H_T) \\ &\geq Tr \mathbb{P}\{\bar{L}_\chi(Y, H_T) > r\} \\ &= Tr(1 - q_r^{\text{Alg}}), \end{aligned}$$

which proves (64). If (65) holds, then $q_r^{\text{Alg}} \leq 1/2$, giving (66). Finally,

$$\text{kl}\left(\frac{1}{2}, p\right) = \frac{1}{2} \log \frac{1}{4p(1-p)} \geq \frac{1}{2} \log \frac{1}{p} - \log 2,$$

so (67) implies (65). □

Relation to interactive Fano. Theorem 6.1 is an adaptation of the interactive Fano method of Chen et al. (2024) to the Bellman-sufficient representation framework. The binary-KL step is the same ghost-data argument: one compares the true experiment with a reference experiment, applies data processing to the small-regret event, and obtains a Bayes risk lower bound when the trajectory information is smaller than the logarithmic inverse ghost-good probability. The additional point here is representational. We choose the reference experiment to be the Bayesian posterior-predictive trajectory and choose an information index $Y = \chi(\Omega)$. With this choice, the trajectory information admits the exact stepwise identity through Bellman states:

$$I_\mu(Y; H_T) = \mathbb{E}_{H'_T \sim \bar{\mathbb{P}}_\mu^{\text{Alg}}} \sum_{t=1}^T \mathcal{I}_\chi(S'_t, p'_t).$$

Thus the same reference history H'_T generates both the ghost quantile and the indexed information telescope. This is the lower-bound counterpart of the AIR/MAIR upper-bound identity. In particular, $\chi(\Omega) = \Omega$ gives the model-index MAIR form, while $\chi(\Omega) = \pi^*(\Omega)$ gives the action-index AIR form.

Remark 6.2 (Why average notation is useful). *The theorem could be written entirely with cumulative quantities. Writing $\bar{C}_\chi = T^{-1}\mathbb{E} \sum_t \mathcal{I}_\chi$ and $\bar{L} = T^{-1}L$ makes the scaling visible:*

$$\text{information side} = T\bar{C}_\chi, \quad \text{regret side} = Tr.$$

Thus T appears exactly as in one-round reductions, but no posterior trajectory has been collapsed into a static action distribution.

6.3 Bellman–Fano lower-bound method

Theorem 6.1 is algorithm-specific. We now give an algorithm-uniform lower-bound method. The first bounds the information of any algorithm; the second bounds the ghost probability of success of any algorithm.

Exact information Bellman value and supersolutions. The intrinsic algorithm-uniform information quantity is an exact Bellman value. For a function F on states, define

$$(\mathbf{C}_t F)(s) := \sup_{p \in \Delta(\mathcal{A}_t(s))} \left\{ \mathcal{I}_\chi(s, p) + \mathbb{E}_{a \sim p, o \sim P_{s,a}} F(\tau(s, a, o)) \right\}.$$

Set

$$C_{T+1}^*(s) = 0, \quad C_t^*(s) = (\mathbf{C}_t C_{t+1}^*)(s).$$

Then $C_1^*(s_1)$ is the exact posterior-reference indexed-information capacity on the retained state. A sequence $\bar{C}_t$ is a valid information supersolution if

$$\bar{C}_{T+1} \geq 0, \quad \bar{C}_t(s) \geq (\mathbf{C}_t \bar{C}_{t+1})(s) \quad \forall t, s.$$

For every algorithm starting from s_1 ,

$$C_\chi^{\text{Alg}}(\mu) \leq C_1^*(s_1) \leq \bar{C}_1(s_1). \tag{68}$$

Thus the exact Bellman recursion is the tight intrinsic object; supersolutions are used only as computable or analytically convenient upper bounds.

Exact ghost-good Bellman value and supersolutions. Let $m \in \Delta(\mathcal{Y})$ be the target-index distribution used to evaluate success, let s be the posterior-reference state, and let $g : \mathcal{Y} \rightarrow \mathbb{R}$ be an accumulated average-loss profile. For threshold r , define the exact ghost-good Bellman value by

$$\Gamma_{T+1}^{r,*}(m, s, g) = m\{y : g(y) \leq r\},$$

$$\Gamma_t^{r,*}(m, s, g) = \sup_{p \in \Delta(\mathcal{A}_t(s))} \mathbb{E}_{a \sim p, o \sim P_{s,a}} \Gamma_{t+1}^{r,*} \left(m, \tau(s, a, o), g + \frac{1}{T} \ell^X(s, a) \right),$$

where $\ell^X(s, a)$ is the function $y \mapsto \ell_y^X(s, a)$. A function sequence $\bar{\Gamma}_t^r$ is a valid ghost-mass supersolution if it dominates the terminal condition and the same Bellman right-hand side with $\bar{\Gamma}_{t+1}^r$ in place of $\Gamma_{t+1}^{r,*}$. Then, for every algorithm,

$$p_r^{\text{Alg}}(\mu, \chi) \leq \Gamma_1^{r,*}(\mu_\chi, s_1, 0) \leq \bar{\Gamma}_1^r(\mu_\chi, s_1, 0). \quad (69)$$

The exact ghost entropy and a certified ghost entropy are therefore

$$\mathcal{E}_{*,1}^r(\mu, \chi) := -\log \Gamma_1^{r,*}(\mu_\chi, s_1, 0), \quad \underline{\mathcal{E}}_1^r(\mu, \chi) := -\log \bar{\Gamma}_1^r(\mu_\chi, s_1, 0). \quad (70)$$

The certified entropy is conservative: $\underline{\mathcal{E}}_1^r \leq \mathcal{E}_{*,1}^r$.

Theorem 6.3 (Bellman–Fano quantile-index lower bound). *Assume the cumulative indexed loss is nonnegative as in Theorem 6.1. Suppose $\bar{C}_t$ is an information supersolution and $\bar{\Gamma}_t^r$ is a ghost-good-mass supersolution; the exact choices $\bar{C} = C^*$ and $\bar{\Gamma} = \Gamma^{r,*}$ give the tight intrinsic Bellman–Fano values. The statement is for the unrestricted nonanticipating class; for a Bellman-compatible restricted class, replace each local supremum in the exact recursions and in the supersolution recursions by the corresponding local feasible set. Then*

$$\inf_{\text{Alg} \in \mathfrak{A}_T^{\text{na}}} \mathbb{E}_{\Omega \sim \mu} \mathbb{E}_{\mathbb{P}_\Omega^{\text{Alg}}} L_\Omega(H_T) \geq Tr \left[1 - \psi(e^{-\underline{\mathcal{E}}_1^r(\mu, \chi)}, \bar{C}_1(s_1)) \right]. \quad (71)$$

In particular, if

$$\bar{C}_1(s_1) + \log 2 \leq \frac{1}{2} \underline{\mathcal{E}}_1^r(\mu, \chi), \quad (72)$$

then

$$\inf_{\text{Alg} \in \mathfrak{A}_T^{\text{na}}} \mathbb{E}_{\Omega \sim \mu} \mathbb{E}_{\mathbb{P}_\Omega^{\text{Alg}}} L_\Omega(H_T) \geq \frac{Tr}{2}. \quad (73)$$

Consequently,

$$\mathfrak{R}_T^* \geq \sup_{\mu, \chi, r} \left\{ \frac{Tr}{2} : \bar{C}_1(s_1) + \log 2 \leq \frac{1}{2} \underline{\mathcal{E}}_1^r(\mu, \chi) \right\}.$$

Proof. For every algorithm, (68) gives $T\bar{C}_\chi^{\text{Alg}}(\mu) = C_\chi^{\text{Alg}}(\mu) \leq \bar{C}_1(s_1)$, and (69) gives $p_r^{\text{Alg}}(\mu, \chi) \leq e^{-\underline{\mathcal{E}}_1^r(\mu, \chi)}$. Substitute these two inequalities into Theorem 6.1, using the monotonicity of ψ in both arguments. The minimax statement follows from Yao’s principle. $\square$

The critical average-regret radius is

$$r_*(\mu, \chi) := \sup \left\{ r : \bar{C}_1(s_1) + \log 2 \leq \frac{1}{2} \underline{\mathcal{E}}_1^r(\mu, \chi) \right\}. \quad (74)$$

The lower bound is order Tr_* . A two-point Le Cam proof corresponds to the special case in which $\underline{\mathcal{E}} = O(1)$. A Fano or local-prior-mass lower bound has $\underline{\mathcal{E}}$ of order dimension or effective dimension.

6.4 DEC as one-step complexity

As discussed in Section 5, offset DEC is obtained from the fixed-truth log-potential bracket (40) by making a one-step relaxation: the coordinate potential drop, or its posterior-averaged Bellman continuation counterpart, is replaced by a static one-step separation penalty, and the posterior-reference law is replaced by a fixed one-step reference predictive law.

In the model-index case $\chi(\omega) = \omega$, let $\mathfrak{C}_t(s)$ be the local comparison set at the current time-state pair and let $\bar{\nu}$ be a reference belief at state s . We write

$$P_{\bar{\nu},s,a} := \int P_{\omega,s,a} \bar{\nu}(d\omega)$$

for its predictive law. The corresponding pointwise KL offset is

$$\text{dec}_{\gamma}^{\text{KL}}(s; \bar{\nu}) := \inf_{p \in \Delta(\mathcal{A}_t(s))} \sup_{\omega \in \mathfrak{C}_t(s)} \{ \mathbb{E}_{a \sim p} \ell_{\omega}(s, a) - \gamma \mathbb{E}_{a \sim p} D_{\text{KL}}(P_{\omega,s,a} \| P_{\bar{\nu},s,a}) \}.$$

(one-step model-index offset)

Thus the usual offset DEC may be viewed as a worst-case, static-reference relaxation of the Bellman program: the learner chooses the one-step decision distribution p , and the local adversary then chooses the hardest model relative to the fixed reference predictive law.

The same one-step logic also appears on the lower-bound side, but only after an additional reduction from a dynamic posterior-reference experiment to a one-step reference experiment. Theorem 6.1 and Theorem 6.3 retain the dynamic reference posteriors. A one-step quantile DEC is obtained only by relaxing this dynamic proof: one fixes a reference model, or more generally a reference predictive law, decomposes the trajectory divergence into one-step divergences, and replaces the algorithm-dependent ghost history by a one-round decision distribution. In Chen et al. (2024, Section 3.2.2), this idea is made explicit through the quantile PAC-DEC. A reference model $\bar{M}$ generates the ghost experiment, an adversarial model M supplies the hard alternative, and the loss certificate is a quantile of the one-step risk rather than its expectation. The further reduction to constrained regret-DEC in that paper is precisely where localization, regularity, and problem-specific assumptions enter.

The passage from the dynamic theorem to a one-step coefficient is therefore not merely notational. On both the upper- and lower-bound sides, existing DEC measures may be viewed as one-step relaxations obtained by freezing the posterior-reference process into reference models or predictive laws. This interpretation is primarily conceptual: even the distinction between quantile and expected loss raises nontrivial technical questions when deriving regret-DEC lower bounds under regularity conditions. Nevertheless, the present paper adopts this viewpoint because it clarifies how one-step DEC values arise from the dynamic indexed Bellman picture.

Thus DEC may be read as a one-step relaxation of indexed Bellman information complexity. When the posterior geometry is simple, this relaxation can be sharp. In more structured problems, however, a one-round coefficient may obscure the dynamic object either by double-counting an entropy or estimation cost already accounted for by the posterior-reference lower bound, or by discarding the reference-trajectory geometry needed for localization. This issue appears in several examples: global RKHS DEC can be vacuous without localization, as discussed at the end of Section 7.1, or without a critical-radius regularity condition (Foster et al., 2023); in multi-armed bandits, an E2D upper bound may charge the cardinality complexity twice; in linear bandits, collapsing the adaptive posterior trajectory to a static action law can introduce an artificial gap in the lower bound; and for Bellman rank, DEC-style questions become intrinsic only after the relevant observable witnesses and their estimation complexity are fixed (Jiang et al., 2017; Du et al., 2021). The examples in Section 7 should therefore be viewed as indexed-Bellman certificates and diagnostics for when one-step DEC minimax calculation can be improved.

7 Applications: information-complexity sandwich examples

This section reorganizes the examples around the information-risk sandwich of Section 3. The role of the examples is analogous to the concrete statistical applications in the information-theoretic work of Zhang (2006); Yang and Barron (1999); Yu (1997): an abstract upper–lower information principle becomes meaningful only after the state, index, hard prior, entropy radius, and upper value are all verified in a concrete experiment. In the sequential setting, fixed-sample entropy is replaced by ghost entropy along reference histories, and an estimator is replaced by a Bellman program or a tractable certificate for it. The first kernel-bandit subsection records the upper side and the need for localization. The next two subsections give complete finite-index and hypercube sandwich calculations for Gaussian bandits.

7.1 Finite-action kernel bandits: algorithms and upper bounds

This subsection specializes the indexed Bellman language to kernel bandits on a finite active action set. The point is not to make the unknown function finite-dimensional. The truth remains an element of an RKHS. What becomes finite is the marginal experiment observed through the active actions. This finite marginal is enough to define the GP posterior update, the model-index MAIR information, and the action-index AIR belief over the optimal active action. We present the exact finite-marginal indexed Bellman program first, and then compare GP-UCB, GP-E2D, and AMS/EBO as tractable relaxations of its bracket.

Problem and finite active marginal. Let $\mathcal{H}_k$ be a reproducing kernel Hilbert space (RKHS) on a possibly infinite domain $\mathcal{X}_{\text{all}}$, with kernel k satisfying $k(x, x) \leq \kappa^2$. The learner is evaluated on a finite active set

$$\mathcal{X} = \{x_1, \dots, x_n\} \subset \mathcal{X}_{\text{all}}.$$

The unknown reward function $f^* \in \mathcal{H}_k$ satisfies $\|f^*\|_{\mathcal{H}_k} \leq B$, and at round t the learner chooses $X_t \in \mathcal{X}$ and observes

$$Y_t = f^*(X_t) + \varepsilon_t, \quad \mathbb{E}[\varepsilon_t \mid H_{t-1}, X_t] = 0, \quad (75)$$

where the noise is conditionally R -sub-Gaussian. For the Gaussian calculations below we use the algorithmic reference model with variance parameter $\lambda > 0$; the frequentist confidence event is obtained by the usual self-normalized argument.

Let $F = (f(x_1), \dots, f(x_n)) \in \mathbb{R}^n$ be the active reward vector and let $K_{\mathcal{X}}$ be the $n \times n$ kernel matrix. The GP reference prior is

$$F \sim N(0, K_{\mathcal{X}}), \quad Y_t \mid F, X_t = x_i \sim N(F_i, \lambda).$$

After history H_{t-1} , the algorithmic posterior on F is Gaussian,

$$F \mid H_{t-1} \sim N(m_t, \Sigma_t).$$

For $x_i \in \mathcal{X}$, write $m_t(x_i) = e_i^\top m_t$ and $s_t^2(x_i) = e_i^\top \Sigma_t e_i$. If $X_t = x_i$, the exact posterior update is

$$m_{t+1} = m_t + \frac{\Sigma_t e_i}{\lambda + s_t^2(x_i)} \{Y_t - m_t(x_i)\}, \quad (76)$$

$$\Sigma_{t+1} = \Sigma_t - \frac{\Sigma_t e_i e_i^\top \Sigma_t}{\lambda + s_t^2(x_i)}. \quad (77)$$

This is a finite marginal posterior update. It should not be read as an assumption that f^* has n parameters. The RKHS function may be infinite-dimensional; only the observations and decisions in this finite experiment depend on the vector F .

Model-index information and action-index information. For the model-index specialization, take the index to be the active reward vector $Y_{\text{MAIR}} = F$. The one-step GP/MAIR information at state H_{t-1} is

$$\mathcal{I}_t^{\text{GP}}(p) := \mathbb{E}_{x \sim p} \frac{1}{2} \log \left(1 + \frac{s_t^2(x)}{\lambda} \right). \quad (78)$$

Along a realized action sequence,

$$C_T^{\text{GP}} := \sum_{t=1}^T \mathcal{I}_t^{\text{GP}}(\delta_{X_t}) = \frac{1}{2} \log \det(I + \lambda^{-1} K_{X_{1:T}, X_{1:T}}), \quad (79)$$

where repeated actions are included in the Gram matrix. This is the realized information gain of the algorithmic trajectory. It can be much smaller than the maximal information gain that maximizes over all length- T designs.

For the action-index specialization, assume ties are broken by a fixed rule and set

$$A^*(F) \in \arg \max_{x \in \mathcal{X}} F_x, \quad q_t(a) := \mathbb{P}(A^* = a \mid H_{t-1}).$$

Let ν_t^a be the exact conditional law of F given H_{t-1} and $A^* = a$. It is generally a truncated Gaussian on the cone where a is optimal. Define the conditional predictive law

$$P_{t,a,x}(\cdot) := \int N(F_x, \lambda)(\cdot) \nu_t^a(dF),$$

for the observation at action x given the event $A^* = a$. The marginal predictive is

$$P_{t,x} = \sum_{a \in \mathcal{X}} q_t(a) P_{t,a,x} = N(m_t(x), \lambda + s_t^2(x)).$$

The AIR information increment is

$$\mathcal{I}_t^{\text{AIR}}(p) := \mathbb{E}_{x \sim p} \sum_{a \in \mathcal{X}} q_t(a) D_{\text{KL}}(P_{t,a,x} \| P_{t,x}). \quad (80)$$

The conditional AIR regret is

$$\Delta_t^{\text{AIR}}(p) := \sum_{a \in \mathcal{X}} q_t(a) \mathbb{E}_{F \sim \nu_t^a} \mathbb{E}_{x \sim p} [F_a - F_x]. \quad (81)$$

The exact indexed chain rule gives

$$\sum_{t=1}^T \mathbb{E} \mathcal{I}_t^{\text{AIR}}(p_t) = I(A^*; H_T) \leq H(A^*) \leq \log n.$$

Thus AIR places the entropy telescope on the decision index rather than on the ambient RKHS. The price is that the coefficient converting action-index information into regret must be controlled by an algorithmic bracket.

Algorithm 1: finite-marginal AIR/MAIR Bellman DP. On the finite active marginal, the exact frequentist benchmark is the log-penalized Bellman program of Theorem 5.2. The state contains (m_t, Σ_t, q_t) , where (m_t, Σ_t) is the algorithmic GP marginal on F and q_t is the maintained reference law of the index, either $A^*(F)$ for AIR or F for MAIR. Let $\mathfrak{C}_t(m, \Sigma, q)$ be a calibrated local set of candidate active reward vectors containing the fixed truth F^* on the calibration event. In the action-index version define

$$y(f) = A^*(f), \quad \mathbf{G}_t^{\text{AIR}}(p; f) := \mathbb{E}_{x \sim p, O \sim N(f_x, \lambda)} \log \frac{q^+(y(f) \mid m, \Sigma, q, x, O)}{q(y(f))},$$

where q^+ is the chosen reference update for the optimal-action marginal. The robust finite-marginal Bellman recursion is

$$W_t^\gamma(m, \Sigma, q) = \inf_{p \in \Delta(\mathcal{X})} \sup_{f \in \mathfrak{C}_t(m, \Sigma, q)} \left\{ \Delta_f(p) + \mathbb{E}_f W_{t+1}^\gamma(m^+, \Sigma^+, q^+) - \gamma \mathbf{G}_t^{\text{AIR}}(p; f) \right\}.$$

The model-index MAIR version replaces $y(f)$ by the finite reward vector f and uses the corresponding model-coordinate posterior log gain. By Theorem 5.2, an approximate minimizer of this recursion has regret at most

$$W_1^\gamma(m_1, \Sigma_1, q_1) + \gamma \log \frac{1}{q_1(A^*(F^*))} + \text{optimization and calibration errors}.$$

If the maintained belief is the exact Bayesian posterior and one averages this coordinate recursion over F , the robust supremum is replaced by posterior expectation and the dynamic program reduces to the posterior-averaged AIR recursion

$$U_t^\gamma(m, \Sigma, q) = \inf_{p \in \Delta(\mathcal{X})} \left\{ \Delta_t^{\text{AIR}}(p) + \mathbb{E} U_{t+1}^\gamma(m^+, \Sigma^+, q^+) - \gamma \mathcal{I}_t^{\text{AIR}}(p) \right\}.$$

Thus the finite-marginal Bellman DP is the clean upper object for comparison with the lower theorem: the frequentist version uses the coordinate log potential and calibrated candidate truths, while the Bayesian version is its posterior average. The following three rules are tractable relaxations or certificates of this benchmark.

Algorithm 2: GP-UCB. GP-UCB uses the Gaussian marginal posterior (m_t, Σ_t) on the finite active marginal. In this paragraph, e_x denotes the coordinate vector associated with $x \in \mathcal{X}$, and $s_t(x) = (e_x^\top \Sigma_t e_x)^{1/2}$ denotes the unmultiplied posterior standard deviation, i.e., the generic uncertainty scale $\sigma_t(x)$ from Lemma 5.4. The confidence half-width is

$$w_t(x) := \beta_t s_t(x),$$

where β_t is the confidence multiplier. The algorithm chooses

$$X_t \in \arg \max_{x \in \mathcal{X}} \{m_t(x) + \beta_t s_t(x)\}. \quad (82)$$

The following event is the frequentist calibration:

$$\mathcal{E}_{\text{GP}} := \{|f^*(x) - m_t(x)| \leq w_t(x) = \beta_t s_t(x) \text{ for all } t \leq T, x \in \mathcal{X}\}. \quad (83)$$

For finite $\mathcal{X}$, standard RKHS self-normalized concentration gives $\mathbb{P}(\mathcal{E}_{\text{GP}}) \geq 1 - \delta$ for the usual choice of β_t . For example, up to universal constants one may take

$$\beta_t \asymp B + R \sqrt{\Gamma_{t-1}^{\text{GP}} + \log(1/\delta)}, \quad \Gamma_m^{\text{GP}} := \sup_{x_{1:m} \in \mathcal{X}^m} \frac{1}{2} \log \det(I + \lambda^{-1} K_{x_{1:m}, x_{1:m}}),$$

with the precise form depending on the normalization of k and the ridge parameter. Here $K_{x_{1:m}, x_{1:m}}$ is the Gram matrix of the possibly repeated design sequence $(x_1, \dots, x_m)$. The quantity Γ_m^{GP} is a worst-case calibration device for the confidence event; it is different from the realized information gain C_T^{GP} paid in the regret bound. The next theorem isolates the deterministic part of the GP–UCB proof.

Theorem 7.1 (GP–UCB realized-information bound). *On the event $\mathcal{E}_{\text{GP}}$, the GP–UCB rule (82) satisfies*

$$\text{Reg}_T(f^*) \leq 2\beta_T \sum_{t=1}^T s_t(X_t) \leq 2\beta_T \sqrt{T c_{\kappa, \lambda} C_T^{\text{GP}}}, \quad (84)$$

where

$$c_{\kappa, \lambda} := \frac{2\kappa^2}{\log(1 + \kappa^2/\lambda)}$$

and C_T^{GP} is the realized information gain in (79). Consequently, if rewards are bounded in $[0, 1]$ and $\mathbb{P}(\mathcal{E}_{\text{GP}}) \geq 1 - \delta$, then

$$\mathbb{E} \text{Reg}_T(f^*) \leq 2\beta_T \sqrt{T c_{\kappa, \lambda} \mathbb{E} C_T^{\text{GP}}} + T\delta.$$

Proof. Let $x^* \in \arg \max_{x \in \mathcal{X}} f^*(x)$. On $\mathcal{E}_{\text{GP}}$, calibration at x^* and the optimistic choice of X_t give

$$f^*(x^*) \leq m_t(x^*) + w_t(x^*) \leq m_t(X_t) + w_t(X_t).$$

Calibration at X_t gives $m_t(X_t) \leq f^*(X_t) + w_t(X_t)$. Hence

$$f^*(x^*) - f^*(X_t) \leq 2w_t(X_t) = 2\beta_t s_t(X_t) \leq 2\beta_T s_t(X_t).$$

Summing and applying Cauchy–Schwarz yields

$$\text{Reg}_T(f^*) \leq 2\beta_T \sqrt{T \sum_{t=1}^T s_t^2(X_t)}.$$

For $u \in [0, \kappa^2/\lambda]$, monotonicity of $u/\log(1 + u)$ on a bounded interval gives

$$\lambda u \leq \frac{\kappa^2}{\log(1 + \kappa^2/\lambda)} \log(1 + u).$$

Applying this with $u = s_t^2(X_t)/\lambda$ and using (79) proves (84). The expectation bound adds the trivial regret bound T on $\mathcal{E}_{\text{GP}}^c$ and uses Jensen’s inequality. $\square$

GP–UCB as a MAIR decomposition. The theorem is Lemma 5.4 with the generic uncertainty scale $\sigma_t(X_t) = s_t(X_t)$ and the realized-information log determinant telescope

$$\sum_t \frac{1}{2} \log(1 + s_t^2(X_t)/\lambda) = C_T^{\text{GP}}.$$

The confidence half-width is $w_t(X_t) = \beta_t s_t(X_t)$, but the multiplier β_t is not part of the information telescope; it is the coefficient paid for calibration and optimism in the fixed-truth bracket. This separation is the reason the proof uses both s_t and w_t : s_t^2 is summed by the log-determinant/elliptical-potential certificate, whereas w_t is used only to upper bound instantaneous regret. GP–UCB is often practically strong because it follows the realized trajectory and pays the realized information gain C_T^{GP} , not the worst possible design information Γ_T^{GP} , even though its confidence multiplier may be conservative in step-uniform minimax analyses.

Algorithm 3: GP–E2D. A GP–E2D rule keeps the same GP posterior reference but chooses a distribution over actions by solving a localized DEC offset. Given a localized comparison set $\mathcal{F}_t$ containing plausible functions, define

$$\Delta_f(p) := \max_{x \in \mathcal{X}} f(x) - \mathbb{E}_{x \sim p} f(x),$$

and let $P_{f,x} = N(f(x), \lambda)$ while $P_{t,x} = N(m_t(x), \lambda + s_t^2(x))$ is the posterior predictive. A schematic GP–E2D decision is

$$p_t \in \arg \min_{p \in \Delta(\mathcal{X})} \sup_{f \in \mathcal{F}_t} \{ \Delta_f(p) - \gamma_t \mathbb{E}_{x \sim p} D_{\text{KL}}(P_{f,x} \| P_{t,x}) \}. \quad (85)$$

If $f^* \in \mathcal{F}_t$ and the optimized value in (85) is at most ε_t , then

$$\Delta_{f^*}(p_t) \leq \gamma_t \mathbb{E}_{x \sim p_t} D_{\text{KL}}(P_{f^*,x} \| P_{t,x}) + \varepsilon_t. \quad (86)$$

Summing (86) gives a regret bound once the fixed-truth KL terms are related to a posterior-ratio or information telescope. Under a well-specified Bayesian GP reference this relation holds in expectation by the MAIR chain rule. Under a fixed frequentist truth it requires calibration/localization. This is the technical point that a GP–E2D analysis must address: the one-round DEC is optimized, but the compatibility between that coefficient and the realized GP log determinant is not automatic. An unlocalized offset over an infinite-dimensional RKHS ball can be unbounded or vacuous; meaningful GP–E2D guarantees must localize the comparison set to the posterior-reference experiment.

Algorithm 4: AMS/EBO on the action index. AMS/EBO attacks the same specialized log-potential bracket from the robust posterior side. In the finite-action kernel marginal, the natural AIR index is A^* . Let q_t be the current reference distribution on A^* , let $\mathfrak{B}_t(q_t)$ be a calibrated local set of candidate beliefs ν on the active reward vector F , and fix a constant coefficient $\gamma > 0$. For each ν , define $\Delta_\nu^{\text{AIR}}(p)$ and $\mathcal{I}_\nu^{\text{AIR}}(p)$ by the analogues of (81) and (80), using the conditional laws induced by ν . A robust AIR/EBO decision is

$$p_t \in \arg \min_{p \in \Delta(\mathcal{X})} \sup_{\nu \in \mathfrak{B}_t(q_t)} \{ \Delta_\nu^{\text{AIR}}(p) - \gamma \mathcal{I}_\nu^{\text{AIR}}(p) - \gamma D_{\text{KL}}(\nu_{A^*} \| q_t) \}. \quad (87)$$

The maximization over ν is essential: without it the display is only the posterior AIR bracket for a chosen belief, not the robust AMS/EBO relaxation. In a finite-index abstraction, the last two terms are the KL-dual form of the log-partition potential (45). If the robust value in (87) is at most ε_t , the required concavity/first-order conditions in Lemma 5.5 hold, and the belief set is calibrated so that the fixed truth f^* is an admissible comparison direction up to error cal_T , then Theorem 5.1 gives

$$\mathbb{E}_{f^*} \text{Reg}_T \leq \gamma \log \frac{1}{q_1(a^*)} + \mathbb{E}_{f^*} \sum_{t=1}^T \varepsilon_t + \text{cal}_T, \quad (88)$$

where $a^* = A^*(f^*)$. The constant coefficient avoids the false shortcut of replacing a weighted fixed-truth log-gain sum by a single telescope. If a varying coefficient is used, the corresponding weighted telescope must be proved separately. Under a Bayesian reference prior, posterior averaging of the same coordinate bound gives

$$\mathbb{E} \text{Reg}_T \leq \gamma H(A^*) + \mathbb{E} \sum_{t=1}^T \varepsilon_t + \mathbb{E} \text{cal}_T.$$

A ratio form gives the familiar Cauchy–Schwarz variant after replacing the fixed coefficient by the appropriate sequential AIR ratio and applying the action-index chain rule. Since $H(A^*) \leq \log n$, this route scales with the finite decision index and the sequential AIR coefficient. It is different from GP–E2D: AMS/EBO uses a robust action-index belief trajectory, whereas GP–E2D optimizes a one-round model-index offset whose relationship to the realized log determinant or posterior-ratio telescope must be justified separately.

Position relative to the GP–UCB optimality debate. The original GP–UCB analysis is practical and trajectory-sensitive because it pays the realized information gain (79), or a worst-case relaxation only at the final step (Srinivas et al., 2010). Later work asks whether the remaining suboptimality comes from the algorithm or from the confidence analysis; in particular, Vakili et al. (2021b) ask for sharper online RKHS confidence intervals and sharper finite-action kernel-bandit guarantees. The indexed view separates the issue into four mechanisms. The finite-marginal indexed Bellman DP is the exact information-potential benchmark, but may be computationally demanding. GP–UCB has the strongest realized-information geometry but may have a conservative MAIR coefficient. AMS/EBO can keep the action-index AIR telescope $I(A^*; H_T) \leq \log n$ and therefore has a principled route to finite-decision complexity, provided the robust bracket can be controlled. GP–E2D directly optimizes a DEC coefficient, but an unlocalized DEC can be vacuous and a localized DEC must be matched to the same posterior-reference trajectory. Thus these algorithms should not be compared by a single universal coefficient; they are one exact Bellman benchmark and three relaxations of the same indexed Bellman identity, with different estimation and coefficient mechanisms.

Why a global RKHS DEC can be vacuous without localization. Consider a Gaussian RKHS bandit with noise variance $\sigma^2 > 0$, feature map $\phi(x) \in \mathcal{H}$ satisfying $\|\phi(x)\|_{\mathcal{H}} \leq 1$, and an action set containing points $\{x_j : j \geq 1\}$ with $\phi(x_j) = e_j$, where $\{e_j\}_{j \geq 1}$ is an orthonormal sequence in $\mathcal{H}$. Let the reference model be $f_0 \equiv 0$, and let

$$\mathcal{F}_B = \{f \in \mathcal{H} : \|f\|_{\mathcal{H}} \leq B\}.$$

For any one-round decision distribution p , define

$$q_j := \mathbb{E}_{x \sim p} \langle e_j, \phi(x) \rangle_{\mathcal{H}}^2.$$

By Parseval’s inequality,

$$\sum_{j \geq 1} q_j \leq \mathbb{E}_{x \sim p} \|\phi(x)\|_{\mathcal{H}}^2 \leq 1.$$

Hence $q_j \rightarrow 0$ along a subsequence. For the alternative $f_j = Be_j$, the action x_j is optimal because

$$f_j(x_j) = B, \quad f_j(x) = B \langle e_j, \phi(x) \rangle_{\mathcal{H}} \leq B$$

for every action x . Its expected one-step regret under p is

$$\mathbb{E}_{x \sim p} \Delta_{f_j}(x) = B - B \mathbb{E}_{x \sim p} \langle e_j, \phi(x) \rangle_{\mathcal{H}} \geq B(1 - \sqrt{q_j}),$$

where the last step uses Cauchy–Schwarz. The one-step KL information against f_0 is

$$\begin{aligned} \mathbb{E}_{x \sim p} D_{\text{KL}}(N(f_j(x), \sigma^2) \| N(0, \sigma^2)) &= \frac{1}{2\sigma^2} \mathbb{E}_{x \sim p} f_j(x)^2 \\ &= \frac{B^2}{2\sigma^2} q_j. \end{aligned}$$

Therefore, for every finite offset coefficient γ , one may choose j with

$$\sqrt{q_j} + \frac{\gamma B}{2\sigma^2} q_j \leq \frac{1}{2},$$

and obtain

$$\mathbb{E}_{x \sim p} \Delta_{f_j}(x) - \gamma \mathbb{E}_{x \sim p} D_{\text{KL}}(N(f_j(x), \sigma^2) \| N(0, \sigma^2)) \geq \frac{B}{2}.$$

Since this holds for every p ,

$$\inf_p \sup_{f \in \mathcal{F}_B} \{\mathbb{E}_{x \sim p} \Delta_f(x) - \gamma \mathbb{E}_{x \sim p} D_{\text{KL}}(P_{f,x} \| P_{0,x})\} \geq \frac{B}{2}.$$

The constrained form exhibits the same obstruction at any positive information radius. If the constraint allows alternatives with one-step information at most $\varepsilon > 0$, choose j such that

$$q_j \leq \min \left\{ \frac{1}{16}, \frac{2\sigma^2\varepsilon}{B^2} \right\}.$$

Then the information of f_j against f_0 is at most ε , while its expected regret under p is at least $3B/4$.

Thus a global, unlocalized DEC over the entire RKHS ball cannot certify an effective finite-dimensional rate in this infinite orthogonal example. This is not merely a technical failure of the coefficient: without localization or additional structure, the class contains infinitely many nearly untested possible best-arm directions. A meaningful effective-dimension analysis must therefore localize the comparison set along the posterior/reference trajectory, or otherwise impose spectral decay, compactness, finite active support, or a calibrated confidence restriction.

7.2 Gaussian multi-armed bandits: finite-index sandwich matching

In this subsection, we apply our lower-bound approach to multi-armed bandits (MAB). Consider $K \geq 3$ Gaussian arms with unit noise variance. Let the ambient model space be a class $\Omega_{\text{MAB}} \subseteq \mathbb{R}^K$ of mean vectors, and write a model environment as $\omega = (\theta_\omega(1), \dots, \theta_\omega(K))$. Pulling arm a in environment ω produces an observation distributed as $N(\theta_\omega(a), 1)$. Let

$$A^*(\omega) \in \arg \max_{a \in [K]} \theta_\omega(a)$$

with an arbitrary fixed tie-breaking rule, and define the information index

$$\chi(\omega) = A^*(\omega) \in [K].$$

Thus the random environment is the full mean vector Ω , while the information target used in the lower bound is only

$$Y = \chi(\Omega) = A^*(\Omega).$$

For the lower bound, fix $\Delta > 0$ and use the K -point prior μ_Δ supported on the spike environments

$$\omega^j = \Delta e_j, \quad j \in [K],$$

where e_j is the j th coordinate vector. Equivalently, draw $J \sim \text{Unif}([K])$ and set $\Omega = \omega^J$. On this prior support the optimal arm is unique and

$$Y = \chi(\Omega) = J.$$

The equality $Y = J$ is only a property of this least-favorable prior; it should not be read as identifying the information target with the full model environment. The one-step regret in environment ω^j is

$$\ell_{\omega^j}(a) = \max_b \theta_{\omega^j}(b) - \theta_{\omega^j}(a) = \Delta \mathbb{1}\{a \neq j\}.$$

For brevity write $\ell_j = \ell_{\omega^j}$ on this subfamily.

Lemma 7.2 (MAB ghost entropy). *Let $Y = \chi(\Omega)$ under the prior $\Omega \sim \mu_\Delta$. For every algorithm and every reference history H'_T independent of $Y \sim \text{Unif}([K])$,*

$$\mathbb{P}\left(\bar{L}_Y(H'_T) \leq \frac{\Delta}{2}\right) \leq \frac{2}{K},$$

where $\bar{L}_y(H'_T) = T^{-1} \sum_{t=1}^T \ell_y(A'_t)$ and A'_t denotes the arm selected in the reference history at time t .

Proof. For a fixed reference action sequence $a'_1, \dots, a'_T$, let

$$N'_y = \sum_{t=1}^T \mathbb{1}\{a'_t = y\}.$$

The average regret against the spike environment ω^y is

$$\bar{L}_y(H'_T) = \Delta(1 - N'_y/T).$$

The event $\bar{L}_y(H'_T) \leq \Delta/2$ implies $N'_y \geq T/2$. Since Y is uniform and independent of the reference sequence,

$$\mathbb{E}[N'_Y | H'_T] = \frac{1}{K} \sum_{y=1}^K N'_y = \frac{T}{K}.$$

Markov's inequality gives

$$\mathbb{P}(N'_Y \geq T/2 | H'_T) \leq \frac{2}{K}.$$

Averaging over H'_T proves the claim. $\square$

Lemma 7.3 (MAB information bound). *There is a universal constant $c_0 > 0$ such that if $\Delta^2 T/K \leq c_0$, then every adaptive algorithm satisfies*

$$I(Y; H_T) \leq \frac{1}{4} \log(K/2),$$

where $Y = \chi(\Omega)$ and $\Omega \sim \mu_\Delta$.

Proof. Let $\mathbb{P}_0$ be the law of the history under the zero-mean reference model, let $\mathbb{P}_j$ be the law under the spike environment ω^j , and let

$$\bar{\mathbb{P}} = \frac{1}{K} \sum_{j=1}^K \mathbb{P}_j$$

be the marginal law of H_T under the prior μ_Δ . Under this prior, $Y = j$ is equivalent to $\Omega = \omega^j$, but the mutual information is still the action-index information $I(Y; H_T)$, not information about an

ambient full mean vector outside the prior support. Since $\bar{\mathbb{P}} \geq K^{-1}\mathbb{P}_j$, the likelihood ratio $d\mathbb{P}_j/d\bar{\mathbb{P}}$ is bounded by K . The inequality

$$D_{\text{KL}}(P\|Q) \leq (2 + \log K)H^2(P, Q) \quad \text{whenever } Q \geq K^{-1}P,$$

together with the triangle inequality and convexity for squared Hellinger distance, gives

$$I(Y; H_T) = \frac{1}{K} \sum_{j=1}^K D_{\text{KL}}(\mathbb{P}_j\|\bar{\mathbb{P}}) \leq 4(2 + \log K) \frac{1}{K} \sum_{j=1}^K H^2(\mathbb{P}_j, \mathbb{P}_0).$$

Using $H^2(P, Q) \leq D_{\text{KL}}(Q\|P)$ and the adaptive Gaussian KL chain rule under the zero model,

$$\frac{1}{K} \sum_{j=1}^K H^2(\mathbb{P}_j, \mathbb{P}_0) \leq \frac{1}{K} \sum_{j=1}^K D_{\text{KL}}(\mathbb{P}_0\|\mathbb{P}_j) = \frac{\Delta^2}{2K} \mathbb{E}_0 \sum_{t=1}^T \sum_{j=1}^K \mathbb{1}\{A_t = j\} = \frac{\Delta^2 T}{2K}.$$

Therefore

$$I(Y; H_T) \leq 2(2 + \log K) \frac{\Delta^2 T}{K}.$$

Choosing, for example,

$$c_0 \leq \inf_{K \geq 3} \frac{\log(K/2)}{8(2 + \log K)}$$

which is a strictly positive universal constant, proves the displayed bound. $\square$

Theorem 7.4 (MAB lower bound). *For Gaussian K -armed bandits with unit noise variance, over any ambient mean-vector class Ω_{MAB} that contains the spike instances $\{\Delta e_j : j \in [K]\}$ used below,*

$$\mathfrak{R}_T^* \geq c\sqrt{KT}$$

for a universal constant $c > 0$ whenever $T \geq K \geq 3$. In particular, this applies to any bounded class such as $[0, 1]^K$ once the universal constant in the choice of Δ is small enough.

Proof. Choose

$$\Delta = c_1 \sqrt{K/T}$$

with c_1 a sufficiently small universal constant, and let $r = \Delta/2$. Lemma 7.2 gives $p_r \leq 2/K \leq 2/3$. The proof of Lemma 7.3 gives the sharper bound

$$T\bar{C} = I(Y; H_T) \leq 2(2 + \log K)c_1^2.$$

Choose c_1 so small that, for every $K \geq 3$,

$$2(2 + \log K)c_1^2 \leq \text{kl}\left(\frac{3}{4}, \frac{2}{K}\right).$$

This is possible because

$$\inf_{K \geq 3} \frac{\text{kl}\left(\frac{3}{4}, \frac{2}{K}\right)}{2(2 + \log K)} > 0.$$

Since $p_r \leq 2/K \leq 2/3 < 3/4$, the exact quantile bound (64) gives

$$\psi(p_r, T\bar{C}) \leq \frac{3}{4}.$$

Therefore

$$\mathfrak{R}_T^* \geq Tr \left(1 - \frac{3}{4}\right) = \frac{T\Delta}{8} = c\sqrt{KT}.$$

for a universal constant $c > 0$. $\square$

Proposition 7.5 (MAB upper bound and sandwich matching). *For Gaussian K -armed bandits with unit noise variance and mean vectors in a bounded cube, there exists a universal constant $C > 0$ and a nonanticipating algorithm, for example a minimax-optimal distribution-free bandit algorithm such as MOSS, whose regret satisfies*

$$\sup_{\omega} \mathbb{E}_{\omega}^{\text{Alg}} L_{\omega}(H_T) \leq C\sqrt{KT}$$

for all $T \geq K \geq 3$, after adjusting constants for the bounded mean range. Consequently, together with Theorem 7.4,

$$\mathfrak{R}_T^*(\Omega_{\text{MAB}}) = \Theta(\sqrt{KT})$$

on every ambient class containing the spike hard family and contained in a bounded mean cube. In the notation of Section 3, the hard prior has $M = K$ index values, $L_0 = \log K$, and Lemma 7.2 gives

$$E_r = \log(1/p_r) \geq \log(K/2)$$

at radius $r = \Delta/2 \asymp \sqrt{K/T}$. Hence the entropy ratio L_0/E_r is bounded by a universal constant for $K \geq 3$, and the upper and lower values close at the same radius.

Proof. The lower bound is Theorem 7.4. The upper bound is the standard distribution-free minimax upper bound for stochastic multi-armed bandits, achieved by MOSS and recorded in modern bandit texts (Audibert and Bubeck, 2009; Lattimore and Szepesvári, 2020). The entropy comparison follows directly from the explicit hard prior: under the uniform optimal-arm index prior $q_1(j) = 1/K$, the upper coordinate code length is $\log K$, while Lemma 7.2 gives $p_r \leq 2/K$. Therefore $L_0/E_r \leq \log K / \log(K/2)$, which is bounded for $K \geq 3$. $\square$

This is the cleanest finite-index example of the Bellman information-risk sandwich. The full environment is the whole mean vector, but the relevant index is the optimal arm. The hard prior, ghost entropy, information bound, and minimax upper algorithm all operate at the same radius.

7.3 Hypercube linear bandits: high-dimensional indexed sandwich

Consider stochastic linear bandits with action set

$$\mathcal{A} = \{a \in \mathbb{R}^d : \|a\|_2 \leq 1\}.$$

Let the ambient model class be the Euclidean unit parameter ball

$$\Omega_{\text{lin}} = \{\omega \in \mathbb{R}^d : \|\omega\|_2 \leq 1\},$$

and regard an environment $\omega \in \Omega_{\text{lin}}$ as the full mean parameter vector. At time t , after choosing $A_t \in \mathcal{A}$, the learner observes

$$O_t = \langle \omega, A_t \rangle + \xi_t, \quad \xi_t \sim N(0, 1),$$

where the noises are independent over time and independent of the learner's external randomization. For $\omega \neq 0$, the unique optimal action over $\mathcal{A}$ is

$$A^*(\omega) = \frac{\omega}{\|\omega\|_2}.$$

At $\omega = 0$, fix any deterministic tie-breaking action; this case is not used by the lower-bound prior. The information index is the optimal action

$$\chi(\omega) = A^*(\omega), \quad Y = \chi(\Omega),$$

not the full environment Ω . Thus two environments that have the same optimal action are not distinguished by the index unless the prior support itself makes them distinguishable.

We prove the standard $\Omega(d\sqrt{T})$ minimax lower bound through the quantile indexed information theorem. The proof uses a hypercube prior together with an adaptive information bound. The role of the hypercube variable below is only to code the finitely many optimal actions in the prior support. It should not be read as replacing the ambient environment Ω by a model label. Note that for a non-Gaussian prior, one cannot condition on the realized adaptive design matrix and then apply the fixed-design Gaussian-channel capacity formula, because the design itself is a statistic of the past observations and may carry information about the unknown environment and hence about Y . [Chen et al. \(2024, Section 3.3\)](#) prove a closely related lower bound by a different route: they use the interactive Fano method with a Gaussian prior and a unit-sphere calculation. Both approaches show that action-indexed information arguments can recover the sharp linear-bandit order, whereas model-indexed relaxations such as DEC can lose factors when they charge for estimating more of the model than is needed to identify the optimal action.

Let

$$V \sim \text{Unif}(\{\pm 1\}^d), \quad \Omega = \Delta V.$$

Assume $\Delta\sqrt{d} \leq 1$, so that this prior is supported on Ω_{lin} . On this prior support,

$$Y = \chi(\Omega) = A^*(\Omega) = \frac{V}{\sqrt{d}}. \tag{89}$$

The map $v \mapsto v/\sqrt{d}$ is one-to-one on $\{\pm 1\}^d$. Hence, for this prior only, V may be used as a code for the action index Y , and

$$I(Y; H_T) = I(V; H_T).$$

For an index value $y = v/\sqrt{d}$, write ℓ_v for the regret under the corresponding environment $\omega = \Delta v$. The one-step regret of action $a \in \mathbb{R}^d$, $\|a\|_2 \leq 1$, is

$$\ell_v(a) = \Delta\sqrt{d} - \Delta\langle v, a \rangle.$$

Assume $T \geq d^2$ and choose $\Delta \leq 1/\sqrt{d}$, so that $\|\Omega\|_2 = \Delta\sqrt{d} \leq 1$.

Lemma 7.6 (Linear ghost entropy). *There is a universal constant $c_{\text{gh}} > 0$ such that every deterministic reference action sequence $a'_1, \dots, a'_T$, with $\|a'_t\|_2 \leq 1$, satisfies*

$$\mathbb{P}_{V \sim \text{Unif}(\{\pm 1\}^d)} \left(\frac{1}{T} \sum_{t=1}^T (\Delta\sqrt{d} - \Delta\langle V, a'_t \rangle) \leq \frac{\Delta\sqrt{d}}{4} \right) \leq \exp(-c_{\text{gh}}d).$$

Consequently, the same bound holds after averaging over any random reference history H'_T whose induced action sequence is independent of the action index Y , equivalently independent of V under the hypercube prior.

Proof. Let

$$\bar{a} := \frac{1}{T} \sum_{t=1}^T a'_t.$$

Since $\|\bar{a}\|_2 \leq T^{-1} \sum_t \|a'_t\|_2 \leq 1$, the event in the display implies

$$\Delta\sqrt{d} - \Delta\langle V, \bar{a} \rangle \leq \frac{\Delta\sqrt{d}}{4},$$

or equivalently,

$$\langle V, \bar{a} \rangle \geq \frac{3\sqrt{d}}{4}.$$

The random variable $\langle V, \bar{a} \rangle = \sum_{i=1}^d V_i \bar{a}_i$ is a centered Rademacher sum with variance proxy $\|\bar{a}\|_2^2 \leq 1$. Hoeffding's inequality therefore gives

$$\mathbb{P}\left(\langle V, \bar{a} \rangle \geq \frac{3\sqrt{d}}{4}\right) \leq \exp\left(-\frac{9d}{32}\right).$$

Thus the first claim holds, for example with $c_{\text{gh}} = 9/32$. If the reference action sequence is random but independent of Y , then because $Y = V/\sqrt{d}$ is a bijective function of V on the prior support, the reference sequence is independent of V . Conditioning on the reference history and applying the deterministic bound gives the second claim. $\square$

Lemma 7.7 (Adaptive hypercube information capacity). *For the hypercube prior above, every possibly randomized adaptive algorithm satisfies*

$$I(Y; H_T) = I(V; H_T) \leq \Delta^2 T,$$

where $H_T = (A_1, O_1, \dots, A_T, O_T)$.

Proof. Since $Y = V/\sqrt{d}$ is a bijective function of V on the hypercube support, $I(Y; H_T) = I(V; H_T)$. It remains to bound $I(V; H_T)$. Let U denote the learner's internal random seed, independent of V . Then

$$I(V; H_T) \leq I(V; H_T, U) = \mathbb{E}_U I(V; H_T \mid U).$$

It is therefore enough to prove the claim after conditioning on U , so that the algorithm is deterministic.

For $v \in \{\pm 1\}^d$, let $\mathbb{P}_v$ denote the trajectory law when $\Omega = \Delta v$. We use the chain rule over coordinates:

$$I(V; H_T) = \sum_{i=1}^d I(V_i; H_T \mid V_1, \dots, V_{i-1}).$$

Fix a coordinate i and a prefix $u = (v_1, \dots, v_{i-1})$. Let $\mathbb{P}_{u,+}$ and $\mathbb{P}_{u,-}$ be the trajectory laws after averaging uniformly over the remaining coordinates $V_{i+1}, \dots, V_d$, conditional respectively on $V_i = +1$ and $V_i = -1$. Then

$$I(V_i; H_T \mid V_{<i} = u) = D_{\text{JS}}(\mathbb{P}_{u,+}, \mathbb{P}_{u,-}),$$

where D_{JS} denotes Jensen–Shannon divergence with equal weights. For any two probability measures P, Q ,

$$D_{\text{JS}}(P, Q) \leq \frac{1}{4} \left(D_{\text{KL}}(P \parallel Q) + D_{\text{KL}}(Q \parallel P) \right).$$

Next couple the two mixtures by using the same suffix. If $w \in \{\pm 1\}^{d-i}$ and

$$v^+(u, w) = (u, +1, w), \quad v^-(u, w) = (u, -1, w),$$

then joint convexity of relative entropy gives

$$D_{\text{KL}}(\mathbb{P}_{u,+} \parallel \mathbb{P}_{u,-}) \leq \mathbb{E}_w D_{\text{KL}}(\mathbb{P}_{v^+(u,w)} \parallel \mathbb{P}_{v^-(u,w)}),$$

and the same argument gives the analogous reverse inequality.

For two fixed vertices v^+ and v^- that differ only in coordinate i , the adaptive KL chain rule gives

$$\begin{aligned} D_{\text{KL}}(\mathbb{P}_{v^+} \parallel \mathbb{P}_{v^-}) &= \frac{1}{2} \mathbb{E}_{v^+} \sum_{t=1}^T \left(\Delta \langle v^+ - v^-, A_t \rangle \right)^2 \\ &= 2\Delta^2 \mathbb{E}_{v^+} \sum_{t=1}^T A_{t,i}^2. \end{aligned}$$

Here the action kernel contributes no KL because, after conditioning on the learner's internal randomness, the algorithm uses the same decision rule under both environments; only the Gaussian observation kernel changes. Similarly,

$$D_{\text{KL}}(\mathbb{P}_{v^-} \parallel \mathbb{P}_{v^+}) = 2\Delta^2 \mathbb{E}_{v^-} \sum_{t=1}^T A_{t,i}^2.$$

Combining the preceding displays yields

$$\begin{aligned} I(V_i; H_T \mid V_{<i} = u) &\leq \frac{\Delta^2}{2} \mathbb{E}_w \left[\mathbb{E}_{v^+(u,w)} \sum_{t=1}^T A_{t,i}^2 + \mathbb{E}_{v^-(u,w)} \sum_{t=1}^T A_{t,i}^2 \right] \\ &= \Delta^2 \mathbb{E} \left[\sum_{t=1}^T A_{t,i}^2 \mid V_{<i} = u \right], \end{aligned}$$

where the final expectation is under the original hypercube prior and the algorithm's trajectory law. Averaging over $V_{<i}$ and summing over i gives

$$\begin{aligned} I(V; H_T) &\leq \Delta^2 \mathbb{E} \sum_{t=1}^T \sum_{i=1}^d A_{t,i}^2 \\ &= \Delta^2 \mathbb{E} \sum_{t=1}^T \|A_t\|_2^2 \\ &\leq \Delta^2 T. \end{aligned}$$

This proves the claim. □

Remark 7.8 (Why the fixed-design log-determinant argument is not used). *For a Gaussian prior on a full parameter vector, posterior conjugacy gives the adaptive information identity*

$$I(\Theta; H_T) = \frac{1}{2} \mathbb{E} \log \det \left(I + \Sigma_0^{1/2} \left(\sum_{t=1}^T A_t A_t^\top \right) \Sigma_0^{1/2} \right),$$

and this is bounded by a trace-constrained log determinant. For the hypercube prior, however, the posterior is not Gaussian and the realized design matrix is itself a statistic of the observations. Conditioning only on that realized design matrix controls the conditional observation information given the design; it does not by itself control the information carried by the adaptive design. Lemma 7.7 therefore uses a coordinatewise adaptive KL argument to control $I(Y; H_T)$ directly.

Theorem 7.9 (Linear bandit lower bound). *For stochastic linear bandits with action set $\{a \in \mathbb{R}^d : \|a\|_2 \leq 1\}$, unit Gaussian noise, and unit-ball parameter class $\Omega_{\text{lin}} = \{\omega \in \mathbb{R}^d : \|\omega\|_2 \leq 1\}$,*

$$\mathfrak{R}_T^* \geq c d \sqrt{T}$$

for a universal constant $c > 0$, whenever $T \geq d^2$.

Proof. Assume first that d is larger than a sufficiently large universal constant d_0 to be fixed below. Choose

$$\Delta = c_1 \sqrt{\frac{d}{T}},$$

where $c_1 > 0$ is a sufficiently small universal constant. Since $T \geq d^2$, choosing $c_1 \leq 1$ ensures

$$\Delta \sqrt{d} = c_1 \frac{d}{\sqrt{T}} \leq c_1 \leq 1,$$

so the hypercube prior is supported on the unit parameter ball.

Let

$$r = \frac{\Delta \sqrt{d}}{4}.$$

Lemma 7.6 gives the reference ghost-good probability bound

$$p_r \leq \exp(-c_{\text{gh}} d).$$

Lemma 7.7 gives the adaptive action-index information bound

$$T\bar{C} = I(Y; H_T) = I(V; H_T) \leq \Delta^2 T = c_1^2 d.$$

Choose c_1 so small that $c_1^2 \leq c_{\text{gh}}/4$, and then choose d_0 so large that $\log 2 \leq c_{\text{gh}} d/4$ for all $d \geq d_0$. Then, for all $d \geq d_0$,

$$T\bar{C} + \log 2 \leq \frac{1}{2} \log \frac{1}{p_r}.$$

Theorem 6.1, applied to the action index $Y = \chi(\Omega) = A^*(\Omega)$, gives a Bayes regret lower bound under the hypercube prior of at least

$$\frac{Tr}{2} = \frac{T\Delta\sqrt{d}}{8} = \frac{c_1}{8} d \sqrt{T}.$$

Since this prior is supported on Ω_{lin} , the minimax regret over Ω_{lin} is at least this Bayes risk.

It remains to handle the finitely many dimensions $1 \leq d < d_0$. For these dimensions, use the two-point prior $\Omega = \sigma \delta e_1$, where $\sigma \sim \text{Unif}(\{\pm 1\})$, e_1 is the first coordinate vector, and $\delta = (2\sqrt{T})^{-1}$. This prior is also supported on the unit parameter ball. Under $\sigma = +1$, the optimal action is e_1 ; under $\sigma = -1$, the optimal action is $-e_1$. Let $\mathbb{P}_+$ and $\mathbb{P}_-$ be the laws of the history under these two environments, and let $\mathbb{P}_\pm^{t-1}$ denote the law of the history before action t . For any algorithm,

$$\begin{aligned} \frac{1}{2} \mathbb{E}_+ [\delta(1 - A_{t,1})] + \frac{1}{2} \mathbb{E}_- [\delta(1 + A_{t,1})] &= \delta \left(1 - \frac{1}{2} (\mathbb{E}_+ A_{t,1} - \mathbb{E}_- A_{t,1}) \right) \\ &\geq \delta (1 - \text{TV}(\mathbb{P}_+^{t-1}, \mathbb{P}_-^{t-1})). \end{aligned}$$

By Pinsker's inequality and the adaptive Gaussian KL chain rule,

$$\text{TV}(\mathbb{P}_+^{t-1}, \mathbb{P}_-^{t-1}) \leq \sqrt{\frac{1}{2} D_{\text{KL}}(\mathbb{P}_+^{t-1} \parallel \mathbb{P}_-^{t-1})} \leq \delta \sqrt{t-1} \leq \frac{1}{2}.$$

Therefore each round has Bayes regret at least $\delta/2$, and the total Bayes regret is at least $T\delta/2 = \sqrt{T}/4$. Since $d < d_0$, this is at least $(4d_0)^{-1}d\sqrt{T}$. Reducing the universal constant c , if necessary, completes the proof for all d . $\square$

Proposition 7.10 (Linear-bandit upper bound and near matching). *For stochastic linear bandits on the Euclidean unit ball with unit Gaussian noise and unit-ball parameter class, there are algorithms based on confidence ellipsoids, such as OFUL, with regret*

$$\sup_{\omega \in \Omega_{\text{lin}}} \mathbb{E}_{\omega}^{\text{Alg}} L_{\omega}(H_T) \leq Cd\sqrt{T} \text{polylog}(T, d)$$

for a universal constant C and all $T \geq d$. Consequently Theorem 7.9 is matched in the polynomial order $d\sqrt{T}$, with only logarithmic factors belonging to the particular tractable upper bound. On the hard hypercube prior, the index support has $M = 2^d$, the initial code length is $L_0 = d \log 2$, and Lemma 7.6 gives $E_r \geq c_{\text{gh}}d$ at radius $r = \Delta\sqrt{d}/4 \asymp d/\sqrt{T}$. Thus the Bellman–Fano entropy gap is constant; any remaining gap is an upper-bound or logarithmic-confidence gap rather than an index-entropy gap.

Proof. The upper bound is the standard finite-time guarantee for self-normalized confidence-ellipsoid algorithms in stochastic linear bandits (Abbasi-Yadkori et al., 2011; Lattimore and Szepesvári, 2020). The lower bound and entropy calculation are exactly the proof of Theorem 7.9: the hard index is $Y = V/\sqrt{d}$ with $V \in \{\pm 1\}^d$, so $L_0 = d \log 2$, while Lemma 7.6 yields $p_r \leq e^{-c_{\text{gh}}d}$. $\square$

This example is the high-dimensional analogue of the finite-arm calculation. The hypercube prior is not a model-identification target; it is a finite action-index packing for the optimal direction. The lower proof shows that ghost entropy and information capacity meet at the scale $d\sqrt{T}$. Standard tractable upper algorithms achieve the same scale up to logarithmic terms; an exact log-potential Bellman upper certificate without those logarithms would be the corresponding sharp sandwich theorem.

8 Conclusion

This paper is organized around a single principle: the statistical difficulty of an interactive decision problem is determined only after one has chosen the right sufficient state and the right index. The environment remains the fixed frequentist truth that generates observations and losses. The Bellman-sufficient state is the part of the history on which prediction, loss evaluation, updating, and future control can be continued. The index $Y = \chi(\Omega)$ is the object whose identification is charged. It may be the optimal action, the optimal policy, a value object, an active finite marginal, or the full model; choosing it is part of the representation. This separation is the main conceptual point. It prevents an upper bound from paying for irrelevant model estimation, and it prevents a lower bound from proving hardness for information that the decision problem never requires.

On the upper side, the basic object is not a one-step coefficient, but a Bellman supersolution with a logarithmic information coordinate. For a fixed truth ω^* and $y^* = \chi(\omega^*)$, the coordinate potential

$$\gamma \log \frac{1}{q_t(y^*)}$$

turns indexed learning into a dynamic program: regret is controlled by immediate loss, continuation value, and the fixed-truth log gain of the maintained reference marginal. The exact log-penalized

Bellman program is therefore the canonical information-theoretic upper algorithm. Its posterior-averaged form recovers Bayesian AIR/MAIR identities, but the fixed-truth form is more fundamental for minimax analysis. UCB, E2D, and AMS/EBO are best understood as tractable ways to certify or relax this same Bellman bracket. UCB uses calibration and optimism to upper-bound the fixed-truth bracket; E2D keeps the immediate loss and a one-round separation penalty while dropping or freezing the continuation value; AMS/EBO replaces the exact continuation value by a KL-dual log-partition, or admissible-relaxation, potential over calibrated candidate beliefs.

On the lower side, the same representation supports a Bellman–Fano value comparison. The information term is the exact or certified indexed-information Bellman capacity along posterior-reference trajectories, while the entropy term is the exact or certified ghost-good mass of reference histories whose target-wise regret is small. Thus the lower bound does not ask how many environments can be distinguished in the full model class. It asks how much indexed information the interactive experiment can reveal before it would make too many low-regret ghost histories possible. Theorem 3.8 expresses the resulting matching principle: when the log-penalized Bellman upper value and the Bellman–Fano ghost-quantile lower value close at the same radius on the same state/index representation, the minimax rate is determined at that radius. In this sense, the paper gives an interactive analogue of information-risk matching in fixed statistical experiments: local prior mass is replaced by ghost-good mass, and fixed-design KL is replaced by a controlled Bellman information telescope.

Several problems remain open. One is to identify, for broad classes of interactive problems, minimal or near-minimal Bellman-sufficient states. Another is to understand when convex belief relaxations such as AMS/EBO are tight approximations to the exact log-penalized Bellman program, rather than merely safe relaxations. A third is to develop lower bounds whose ghost-good entropy is as constructive as the corresponding upper algorithm. A fourth direction is to extend the same state/index discipline to settings where exact sufficiency is generally unavailable, so that approximation error must be carried explicitly through every step of the Bellman-recursive analysis. Together, these developments should be especially important for long-horizon and across-episode reinforcement learning, planning, and reasoning, which are beyond the reach of much of the current theory.

The conclusion is therefore methodological as much as technical. To analyze an interactive learning problem, one should not begin by asking for the dimension of the model, or for a universal conversion mechanism between estimation and decision. One should ask: what state representation makes the Bellman recursion close, what index is actually worth learning, what logarithmic potential pays for that index, and what ghost quantile witnesses the matching lower entropy? When these four objects align, learning becomes a special dynamic program whose value coordinate is information itself. The statistical complexity of the problem is then no longer a property of the raw model class alone; it is a property of the Bellman-sufficient representation through which the learner, the upper bound, and the lower bound all see the problem.

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